Fixed Income

A Unified Framework for Pricing, Term Structure, Risk, and Markets

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From no-arbitrage and cash-flow valuation to stochastic pricing, risk sensitivities, portfolio hedging, and market structure.

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The Organizing Principle

Fixed income studies how financial claims promise cash across dates and states; how the term structure determines the present value of those cash flows; how interest-rate uncertainty changes their valuation; and how securities transfer, shape, and hedge the resulting risks.

$$\boxed{\text{No Arbitrage} \longrightarrow \text{Discounting} \longrightarrow \text{Term Structure} \longrightarrow \text{Rate Dynamics}}$$

$$\boxed{\text{Pricing Engine} \longrightarrow \text{Products} \longrightarrow \text{Risk} \longrightarrow \text{Hedging} \longrightarrow \text{Markets}}$$

Each part proceeds from an economic problem to a pricing or sensitivity object, its mathematical logic, and its application. Throughout, $d(t)$ denotes a deterministic time-zero discount factor and D_n a stochastic discount process. These are different but related objects.

Notation and conventions

Unless otherwise specified, F is face value; q is an annual coupon rate; y is annualized semiannually compounded YTM; P is price; R_n is the one-period short rate; $B_{n,m}$ is the time-n value of one currency unit paid at m; $\widetilde{\mathbb{P}}$ denotes the pricing measure; and $\widetilde{\mathbb{E}}_n$ is its time-n conditional expectation. In sensitivity sections, D denotes modified duration, D_{Mac} Macaulay duration, and C convexity. Cash flows are default-free except where credit is explicitly discussed.

Part I — Foundations: No-Arbitrage and the Value of Cash Flows

All subsequent valuation rests on one condition: the same dated, state-contingent payments cannot have inconsistent prices. This turns promises of future cash into tradable building blocks.

Market idealizations and the pricing question

Begin with default-free claims in a frictionless market. Rates and products with uncertain payments will be introduced only after the deterministic pricing logic is established.

Baseline market assumptions

- Time is measured in years; present time is t = 0.
- No default unless stated otherwise.
- Ignore transaction costs and bid–ask spreads; arbitrary fractional long/short positions are allowed; order size does not affect price. This is an ideal frictionless, liquid market.

Law of one price, replication, and arbitrage

If two trades deliver identical cash flows at every relevant date and state, their prices must agree. A price discrepancy creates a zero-net-cost long–short position.

Law of one price

The law of one price says that two securities/portfolios with exactly the same future payments must have the same current price. **Price today of F paid at t:**

$$Fd(t)$$

Deterministic cash-flow price:

$$P = \sum_i F_i d(t_i)$$

Arbitrage

- Arbitrage (strict definition): zero input of capital, zero probability of loss, and strictly positive probability of strictly positive profit.
- Strong arbitrage: the positive profit is certain (probability 1). Every strong arbitrage is an arbitrage.
- Logical implications: no arbitrage ⇒ no strong arbitrage ⇒ law of one price. Converses need not hold in general models.
- Practitioners may use “arbitrage” loosely for high-probability/low-risk trades. under the present conventions, loss probability must truly be zero unless explicitly stated otherwise.

Discount factors: the fundamental price objects

Write $d(t)$ for the price today of one unit of default-free money delivered at t. Unlike a quoted yield, $d(t)$ does not require selecting a compounding convention.

Discount factors and time value of money

- d(t) = time-0 market price of receiving $1 at time t with no default risk. It is an intrinsic price object: no compounding convention is needed to define it.

- No-arbitrage requires d(t)>0. Usually d(t)<1 for t>0 and d(t) decreases with t. d(t)>1 or an increase in d(t) over maturity corresponds to negative interest rates for some interval.

$$\boxed{P_0 = \sum_{i=1}^{N} CF_i \, d(t_i).}$$

Economic intuition. A cash flow arriving later cannot be added directly to one arriving sooner. Each is first converted into an equivalent amount of money today, after which those present values add.

Accumulation and compounding conventions

Compounding is a representation of a discount factor, not an additional pricing principle. The economic price is unchanged when its quoted rate convention changes.

Semiannual compounding

U.S. bond-market convention often uses an annualized rate r compounded twice per year.

$$w = x\left(1 + \frac{r}{2}\right)^{2T}$$

$$r = 2\left[\left(\frac{w}{x}\right)^{1/(2T)} - 1\right]$$

With this convention in the deterministic-cash-flow sections, T is normally a multiple of 0.5 years.

For an annualized nominal rate $r^{(m)}$ compounded m times per year, an investment over t years accumulates by $\left(1 + r^{(m)}/m\right)^{mt}$. Hence the same discount factor can be quoted as

$$d(t) = \left(1 + \frac{r^{(m)}(t)}{m}\right)^{-mt} = e^{-r_c(t)t}.$$

Annual, semiannual, monthly, and continuous quotations are equivalent when their implied $d(t)$ agrees. This extension merely expresses the compounding identities below in generic frequency notation.

Simple interest, clean/dirty prices, accrued interest, and bill-yield conventions

Market payment dates and day-count rules determine the actual invoiced amount. A quoted yield without its convention is not a complete price specification.

Flat/clean versus full/dirty price

- Between coupon dates, the economic/full value of a coupon bond evolves continuously, but the bond value would mechanically drop by one coupon immediately after payment. Market quotation separates price into flat price and accrued interest (AI).

$$P_{\text{full}} = P_{\text{flat}} + AI = \text{PV(future payments)}$$

- Full/invoice/dirty price is market-determined. AI is defined by convention. Flat/clean price = full − AI and is the quoted bond price.
- AI is 0 just after a coupon and equals the full coupon amount immediately before the next coupon.

Simple interest and day-count conventions

Simple interest:

$$\text{Principal} \times \text{Rate} \times \text{Time}$$

- Actual/360: n days count as n/360 of a year. Actual/actual: n days count as n/365 in a non-leap year. Some consumer loan conventions treat each month as 1/12 year.
- Illustration, 5% on $50,000 for 365 days: actual/360 interest=$2,534.72 versus actual/actual=$2,500, a $34.72 difference.

  On $500mm, difference=$347,222.22.

Treasury accrued interest and YTM between coupons

$$AI = \eta F \frac{q}{2}, \qquad \eta = \text{fraction of coupon period elapsed}$$

U.S. Treasury actual/actual:

$$AI = \frac{n}{N} F \frac{q}{2}$$

- N = actual days from last coupon to next; n = days from last coupon to settlement. Example: F=$10,000, q=5.5%, N=182, n=77 ⇒ AI=$116.35.

Let τ be the fraction of a coupon period until the next coupon (τ=1−η), and k the number of additional coupons after the next coupon:

$$P_{\text{full}} = F\lambda^{k+\tau} + F\frac{q}{2}\sum_{i=0}^{k}\lambda^{\tau+i},$$

$$\lambda = \left(1 + \frac{y}{2}\right)^{-1}.$$

- There are k+1 future coupons. Each is discounted with $(1 + y/2)^{-2t}$; here t need not be a multiple of 0.5 years.

Current yield between coupons:

$$\frac{qF}{P_{\text{flat}}}$$

- Actual coupon-payment dates can shift for holidays, creating convention issues in interpreting YTM when cash-flow intervals are not exactly six months.

T-bill discount yield and bond-equivalent yield

$$P = 100\left(1 - \frac{n y_d}{360}\right) \quad \Longrightarrow \quad y_d = \frac{(100 - P)360}{100n}$$

- Discount yield is unusual because it divides the dollar gain by face value 100 rather than purchase price P and assumes a 360-day year.

For n ≤ 182:

$$y_{\text{be}} = \frac{(100 - P)365}{Pn}$$

For n>182, the framework uses the semiannual/coupon-equivalent convention determined by:

$$P\left(1 + \frac{y_{\text{be}}}{2}\right) + \frac{y_{\text{be}}}{365}\left(n - \frac{365}{2}\right)\left(1 + \frac{y_{\text{be}}}{2}\right)P = 100$$

- Bond-equivalent yield is also called coupon-equivalent yield. Day-count/compounding/yield conventions are a “snake pit”: small convention differences can create large dollar differences on large trades.

Part II — The Term Structure: Spot, Forward, Par, and Yield-to-Maturity

Once dated cash can be priced, the entire sequence of discount factors forms the term structure. Different yield quotations are distinct ways of describing or summarizing those prices.

Spot-rate curve and discount-factor conversions

Spot-rate conventions and discount-factor conversions

Semiannual spot:

$$\hat{r}(t) = 2\left[\left(\frac{1}{d(t)}\right)^{1/(2t)} - 1\right]$$

Effective spot:

$$\hat{R}(t) = \left(\frac{1}{d(t)}\right)^{1/t} - 1$$

Continuous spot:

$$r_c(t) = -\frac{1}{t}\ln d(t)$$

$$d(t) = \left(1 + \frac{\hat{r}(t)}{2}\right)^{-2t} = (1 + \hat{R}(t))^{-t} = \exp[-t r_c(t)]$$

- Effective compounding is especially convenient when cash-flow dates are not uniformly spaced; t need not be a multiple of 0.5.
- The graph of $\hat{r}(t)$ is the spot-rate curve / zero-coupon yield curve. Analogous curves use effective or continuous rates.
- For a fixed d(t)>0: $r_c(t) \le \hat{r}(t) \le \hat{R}(t)$, with strict inequalities unless d(t)=1.

t	d(t)	$\hat{r}(t)$ (semiannual)	$\hat{R}(t)$ (effective)
0.5	0.9753	5.065%	5.129%
1.0	0.9506	5.131%	5.197%
1.5	0.9228	5.429%	5.5022%

Illustration illustrates that effective rates exceed the corresponding semiannual-compounded annualized rates; the spot curve happened to be upward sloping, but need not be.

Forward lending from replication

A forward rate is fixed now for borrowing or lending later. It is the break-even reinvestment rate implied by today's market prices, not the unknown spot rate that will actually prevail later.

Forward loans and forward rates

- A forward loan is agreed at time τ, initiated at η, and settled at T (τ≤η<T). The amount/interest rate are fixed at agreement; entering the agreement costs zero.
- Effective forward rate $R^{\text{for}}(\tau, \eta, T)$: per $1 borrowed at η, repayment at T is $(1 + R^{\text{for}})^{T-\eta}$. The rate is known at τ, not before.

At time 0, replicate a forward borrowing/lending arrangement using ZCBs: long a ZCB paying \$1 at η; short a T-maturity ZCB with face A_T. Zero initial cost gives d(η)−A_T d(T)=0.

$$A_T = \frac{d(\eta)}{d(T)}$$

$$\left(1 + R^{\text{for}}(0, \eta, T)\right)^{T-\eta} = \frac{d(\eta)}{d(T)}$$

$$= \frac{(1 + \hat{R}(T))^T}{(1 + \hat{R}(\eta))^\eta}.$$

Conceptual caution: : Understand the replication, not only a memorized formula. Practitioners may use the third time variable as maturity or as loan length, and compounding conventions differ.

For a semiannually compounded forward $r^{\text{for}}(\tau, \eta, T)$:

$$(1 + R^{\text{for}})^{T-\eta} = \left(1 + \frac{r^{\text{for}}}{2}\right)^{2(T-\eta)}$$

- Notation f(t): semiannually compounded forward rate set at time 0 for the six-month loan [t−0.5,t]. Thus f(0.5)=$\hat{r}(0.5)$.

$$\left(1 + \frac{\hat{r}(t)}{2}\right)^{2t} = \prod_{j=1}^{2t}\left(1 + \frac{f(j/2)}{2}\right)$$

$$\left(1 + \frac{\hat{r}(t)}{2}\right)^{2t} = \left(1 + \frac{\hat{r}(t-0.5)}{2}\right)^{2t-1}\left(1 + \frac{f(t)}{2}\right)$$

Forward	Illustration
$f(0.5)$	5.065%
$f(1.0)$	5.197%
$f(1.5)$	6.026%

- Important observation: if the spot curve is upward sloping at t ($\hat{r}(t)$>$\hat{r}(t-0.5)$), then f(t)>$\hat{r}(t)$. If the spot curve is downward sloping, f(t)<$\hat{r}(t)$.

Forward-curve shape and reinvestment implications

Price versus maturity for same-coupon bonds

Two bonds have the same F and q>0; Bond 1 matures at T, Bond 2 at T+0.5. Long Bond 2 and short Bond 1 leaves −F at T and +F(1+q/2) at T+0.5. Compare this with investing F from T to T+0.5 at the forward f(T+0.5).

- If q > f(T+0.5), the longer Bond 2 has the higher time-0 price.
- If q < f(T+0.5), the shorter Bond 1 has the higher time-0 price.

Forward rates are not forecasts of realized future spot rates

- Investor A buys a 6-month zero, then at t=0.5 reinvests for the next 6 months at the then-prevailing spot rate $r_{0.5,1}$. Investor B buys a 1-year zero now.
- B's one-year accumulation can be written with today's f(1); A's second-period accumulation uses the future realized $r_{0.5,1}$. At t=0 there is no way to know whether $r_{0.5,1}$ will be greater than, less than, or equal to f(1).

Coupon reinvestment scenario

- Investor A rolls 6-month ZCBs through T. Investor B buys a coupon bond maturing at T and reinvests each coupon in rolling 6-month zeros.
- If all today's forward rates f(t) exceed the corresponding realized future six-month spot rates $r_{t-0.5,t}$, B ends with more at T. If all forwards are lower than realized future spots, A ends with more.

ZCB:

$$P = Fd(T)$$

Annuity:

$$P = A\sum_{i=1}^{mT} d\left(\frac{i}{m}\right)$$

Semiannual coupon bond:

$$P = Fd(T) + F\frac{q}{2}\sum_{i=1}^{2T} d\left(\frac{i}{2}\right)$$

$$f(t) = 2\left[\frac{d(t-0.5)}{d(t)} - 1\right]$$

- Spot rates are known at time 0. Today's forward rates are also known at time 0. Future spot rates that will prevail at future dates are not known at time 0.
- $\hat{r}(t)$ is a geometric-average-type aggregation of the half-year forward rates through t. Numerically, it is usually close to their arithmetic average:

$$\hat{r}\left(\frac{N}{2}\right) \approx \frac{1}{N}\sum_{i=1}^{N} f\left(\frac{i}{2}\right)$$

- Illustration: [5.065%, 5.197%, 6.026%] average to 5.429%, matching $\hat{r}(1.5)$ to the displayed precision.
- If forward rates are monotonically increasing/decreasing/nondecreasing/nonincreasing, spot rates inherit the same monotonicity. Converse is false: a monotone spot curve can coexist with nonmonotone forwards.
- Useful local relation from the Technical note: $\hat{r}(t) \geq \hat{r}(t-0.5)$ if and only if $f(t) \geq \hat{r}(t)$; the same equivalence holds with strict inequalities.

Yield to maturity: a security-level internal rate

The term structure discounts payments date by date. YTM instead selects one rate that reproduces one entire bond price; it therefore depends on that bond's coupon schedule as well as maturity.

Definition and computation

YTM is the single interest rate y that, when used to discount all future cash flows of a security, reproduces its current market price. The default bond convention here is semiannual compounding.

$$P = F\frac{q}{2}\sum_{i=1}^{2T}\left(1+\frac{y}{2}\right)^{-i} + F\left(1+\frac{y}{2}\right)^{-2T}$$

- For nonnegative cash flows, the YTM is unique. It usually must be solved numerically.

Let $\lambda = 1/(1 + y/2)$. Then:

$$P = F\lambda^{2T} + F\frac{q}{2}\sum_{i=1}^{2T}\lambda^i$$

$$P = F\left[\lambda^{2T} + \frac{q}{2}\frac{\lambda(1-\lambda^{2T})}{1-\lambda}\right]$$

$$y = 2\left(\frac{1}{\lambda} - 1\right)$$

What YTM does — and does not — mean

- Higher YTM does not automatically mean a better investment when comparing different coupon bonds, even with the same maturity.
- Even if a bond is held to maturity, YTM is not necessarily "the return" unless one specifies what happens to coupons received before maturity.
- If YTM stays unchanged between two successive coupon dates, that common YTM is the holding-period yield over that coupon interval.
- Negative YTM is theoretically possible; unless specified otherwise, assume $y > 0$.

Conceptual caution: : On a nonflat yield curve, YTM is a property of the whole bond. It is not a maturity-specific discount rate for each individual cash flow.

Price, coupon rate, and YTM — crucial proposition

$$P > F \iff q > y \qquad P = F \iff q = y \qquad P < F \iff q < y$$

Interpretation: premium bond ⇒ coupon rate exceeds YTM; par bond ⇒ coupon=YTM; discount bond ⇒ coupon<YTM. The proof follows by rewriting price so it is linear and increasing in q/y, with value F at q/y=1.

Alternative YTM conventions and IRR

Effective YTM Y:

$$P = \sum_i \frac{F_i}{(1+Y)^{T_i}}$$

Continuously compounded YTM y_c:

$$P = \sum_i F_i \exp(-T_i y_c)$$

- For positive cash flows, effective/continuous YTM is unique. If cash flows change sign, there may be zero, one, or multiple solutions; the terminology distinguishes "IRR" for sign-changing cash-flow problems.

Numerical YTM example and general facts

- Illustration: Bond #3, T=1.5 years, q=10%, P=106.52. Because P>F, y<10%. Solve $106.52 = 105\lambda^3 + 5\lambda^2 + 5\lambda \Rightarrow \lambda = 0.973635 \Rightarrow y = 5.4158\%$.
- A bond's YTM lies between the lowest and highest spot rates corresponding to cash-flow dates up to maturity.
- If a portfolio is long two securities with deterministic nonnegative payments, its YTM lies between the YTMs of the two components.

- If the term structure is flat, the common spot rate equals the YTM.
- Pull to par: as time to maturity shrinks, premium-bond flat prices fall toward par and discount-bond flat prices rise toward par.
- If the spot curve is not flat, do not discount individual bond cash flows using the bond's YTM (except a ZCB, which has only one payment).
- YTM can always be applied to all of that bond's cash flows together to recover its price.
- Even for same-maturity bonds, one bond's YTM cannot price another bond on a nonflat curve unless their coupons are also the same.

Coupon effect, par yields, and distinct yield curves

The par yield is the coupon that makes a new bond worth its face value. For that *particular par bond*, $q = y$; this generally does not make the par yield equal to the same-maturity zero rate.

Coupon effect

- Coupon-bond YTM generally depends on both coupon q and maturity T; one cannot cancel q when equating market price from the term structure to price from one YTM.

For two same-maturity bonds with $q_2 \geq q_1$:

- If spot rates are nondecreasing with maturity, then $y_1 \geq y_2$: the higher-coupon bond has lower/equal YTM.
- If spot rates are nonincreasing, then $y_2 \geq y_1$: the higher-coupon bond has higher/equal YTM.
- The statements remain valid with strict inequalities when the spot-curve inequalities are strict.

Par-coupon bonds, zeros, and yield curves

$$\text{Par-coupon rate } q(T) = \frac{2[1 - d(T)]}{\sum_{i=1}^{2T} d(i/2)}$$

- For a par coupon bond, q=y by the premium/par/discount proposition.
- For a ZCB, YTM equals the spot rate at maturity: y=$\hat{r}(T)$.

Yield curve	Definition
Zero-coupon yield curve	Spot-rate curve: ZCB YTM versus maturity.
Annuity yield curve	YTM of equal-payment semiannual annuities versus maturity; also self-amortizing yield curve.
Par-coupon yield curve	YTM/coupon of par coupon bonds versus maturity.

- Any one of the three curves completely determines the other two.
- Par-coupon yield is always between the zero-coupon and annuity yields. Whether zero is above/below annuity depends on curve shape; if the zero curve is upward sloping everywhere, zero yield is above annuity yield.
- In particular, with an upward-sloping spot curve, $y_a(T) \leq y_{pc}(T) \leq \hat{r}(T)$, and both the annuity-yield curve and par-coupon-yield curve are upward sloping.

- Replication intuition used in the Technical note: a larger-coupon bond equals a smaller-coupon bond plus an annuity. This is the cleanest way to remember why coupon affects YTM on a nonflat curve.

Why coupon matters under an upward-sloping curve. A larger coupon shifts more cash flow to earlier dates, which are discounted at lower spot rates. As a consequence the whole-stream YTM of a higher-coupon, equal-maturity bond can be lower. The par rate is an annuity-weighted transformation of discount factors; the zero rate discounts just the last payment.

Constructing the curve: sequential bootstrapping and smooth interpolation

A coupon bond is a collection of zeros. Market prices of successively longer bonds imply progressively longer discount factors; interpolated or fitted curves extend this logic to unsampled dates.

Bootstrapping discount factors and replication — Illustration

Instrument	Cash flows / price	Implied result
Bond #1	6-month ZCB, price 97.53 per 100 face	$97.53 = 100d(0.5) \Rightarrow d(0.5) = 0.9753$
Bond #2	1-year, 5% coupon, price 99.87	$99.87 = 2.5d(0.5) + 102.5d(1) \Rightarrow d(1) = 0.9506$
Bond #3	1.5-year, 10% coupon, price 106.52	$106.52 = 5d(0.5) + 5d(1) + 105d(1.5) \Rightarrow d(1.5) = 0.9228$

- A 1-year ZCB (Bond #4) trades at 94.93, while the no-arbitrage price implied by d(1) is 95.06, so Bond #4 is cheap. A price above the replication value would be rich.
- Replication: 100 face of Bond #4 can be replicated by long 100/1.025 face of Bond #2 and short 2.5/1.025 face of Bond #1. Replicating cost = 95.06.
- Arbitrage when Bond #4 is at 94.93: buy Bond #4 and short the replicating portfolio. Initial net receipt ≈ 0.1254 (≈0.13) per 100 face; later net cash flows are zero. On $500mm face, the corresponding illustrative profit is $626,829.27, subject in reality to execution/frictions.

Practical yield-curve construction

- In practice, collect reliable observations from liquid instruments and construct a smooth d(t)/spot curve using interpolation. There are usually more unknown curve values than direct pricing equations, so interpolation choice matters.
- Discount factors are smoother than spot rates; forward rates are more jagged than spot rates.
- Instead of exact interpolation, one can use least-squares fitting to a parametric family. Nelson–Siegel is a four-parameter example for continuously compounded yields:

$$y(t) = a + b\left(\frac{1 - e^{-\lambda t}}{\lambda t}\right) + c\left(\frac{1 - e^{-\lambda t}}{\lambda t} - e^{-\lambda t}\right)$$

For semiannual par observations q_n at dates $t_i = i/2$, the recursion follows immediately from $1 = (q_n/2)\sum_{i=1}^{n} d(t_i) + d(t_n)$:

$$\boxed{d(t_n) = \frac{1 - (q_n/2)\sum_{i=1}^{n-1} d(t_i)}{1 + q_n/2}.}$$

The same discount factors then determine spot, forward, par, annuity, and arbitrary cash-flow prices. This is the conceptual reason that those curves are related rather than independent.

Part III — Deterministic Fixed-Income Instruments

Instruments differ first by the schedule of their promised payments. Their values follow from combining the same dated-cash-flow building blocks.

Securities, bond contracts, and economically distinct risks

Securities, bonds, and fixed income

- A security is any tradable financial instrument. Major categories: equity securities, debt securities, and derivative contracts.
- A derivative is a contract whose exchanged payments are determined by underlying variables, often asset prices. Examples include: stock puts/calls, futures, and interest-rate swaps.
- Buying a bond is economically making a loan to the issuer. The purchase price is the principal advanced; absent default, the holder receives interest and repayment of principal.

Loan/bond form	Cash-flow pattern
Zero-coupon bond	One lump-sum payment (principal + interest economically) at maturity.
Coupon bond	Periodic interest-only coupons; principal plus final coupon at maturity.
Annuity / self-amortizing loan	Periodic level payments; each payment partly interest on outstanding principal and partly principal repayment.

- Literal “fixed-income security”: future payment dates and amounts are known with certainty at issuance. Market usage is broader and includes derivatives on debt securities whose payments can be uncertain; “interest-rate product” would be more descriptive.
- Known future payments are called deterministic cash flows or fixed payments. Many U.S. Treasuries have deterministic cash flows.
- Default/credit/counterparty risk: promised payments may not be made. Unless stated otherwise, this treatment assumes no default.

Zero-coupon bonds, coupon bonds, annuities, and perpetuities

Zero-coupon bonds, coupon bonds, annuities

- ZCB: face value F, maturity T, one payment F at T. ZCBs are the building blocks: any deterministic cash-flow security can be represented as a portfolio of ZCBs. Synonyms: pure discount bond, zero, ZCB.
- Coupon bond: face F, maturity T, annual coupon rate q, m coupon payments/year. Coupon at i/m is Fq/m, i = 1,…,mT; final payment is F(1+q/m). In U.S. bonds, m=2 is the default convention under these conventions.
- Face value is also par value. Above/equal/below face = above/at/below par; “premium to par” and “discount to par” are common terms. Coupon rates at issuance are usually chosen so the initial price is close to par.
- Bond prices are commonly quoted per $100 face unless stated otherwise. For previously issued bonds, unless stated otherwise, assume a coupon has just been paid and the next coupon is exactly 1/m years away. Between coupon dates, accrued interest is separate from the quoted price.
- Annuity: fixed amount A paid m times/year through maturity T. A perpetuity has T=∞. Lifetime annuities and variable-payment annuities also exist, but their maturity/payment amounts are not deterministic in the same simple sense.

Annuity and perpetuity yields

Semiannual annuity:

$$P = A\sum_{i=1}^{2T}\lambda^i = A\frac{\lambda}{1-\lambda}(1-\lambda^{2T})$$

Semiannual perpetuity:

$$P = \frac{2A}{y} \iff y = \frac{2A}{P}$$

m-payments/year perpetuity:

$$P = \frac{mA}{y}$$

- An annuity's YTM is independent of payment size A; it generally depends on maturity T.
- For a very long finite semiannual annuity: $y \approx 2/[\sum_{i=1}^{2T} d(i/2)]$.

For a face-F zero, $P = Fd(T)$. For a semiannual coupon bond and an equal-payment annuity,

$$P_{\text{coupon}} = Fd(T) + \frac{Fq}{2}\sum_{i=1}^{2T} d(i/2), \qquad P_{\text{annuity}} = A\sum_{i=1}^{2T} d(i/2).$$

A par bond prices at F; a premium or discount bond prices above or below F. A perpetuity is the infinite-maturity limit of its annuity when discounting makes the infinite sum converge.

The pricing principle behind floating and inflation-indexed claims

Unlike a plain coupon bond, floaters and inflation-linked securities have cash flows tied to reference rates or an index. Their contractual descriptions belong alongside basic bonds, while exact valuation of rate-contingent payments uses the stochastic machinery developed below.

Part IV — Interest-Rate Uncertainty and Stochastic Discounting

A current term structure fixes today's prices but does not determine tomorrow's interest rates. To value claims whose cash flows depend on future states, pricing must extend from dated cash to dated and state-contingent cash.

One-period replication and risk-neutral probabilities

- Two dates 0 and 1. ZCB price $B_{0,1}$ pays \$1 at 1; define one-period R_0 by $B_{0,1} = 1/(1+R_0)$.
- Risky traded asset: $S_1(H) = uS_0$, $S_1(T) = dS_0$, both positive and distinct. H/T states each have positive physical probability.
- Trading strategy (X_0, Δ_0): total initial capital X_0; Δ_0 shares risky. Risk-free investment is $X_0 - \Delta_0 S_0$. Terminal capital:

$$X_1(\omega) = (X_0 - \Delta_0 S_0)(1+R_0) + \Delta_0 S_1(\omega)$$

No-arbitrage, replication, and risk-neutral probabilities

$$\text{No arbitrage} \iff d < 1 + R_0 < u$$

For payoff $V_1(H), V_1(T)$, solve the two-state replication equations:

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)}$$

$$\tilde{p} = \frac{1 + R_0 - d}{u - d}, \qquad \tilde{q} = \frac{u - 1 - R_0}{u - d}, \qquad \tilde{p} + \tilde{q} = 1$$

$$V_0 = \frac{\tilde{p}\, V_1(H) + \tilde{q}\, V_1(T)}{1 + R_0} = \tilde{\mathbb{E}}\left[\frac{V_1}{1+R_0}\right]$$

- Under no-arbitrage, $\tilde{p}, \tilde{q}$ are strictly positive and define the risk-neutral/pricing measure. The initial capital of the replicating strategy is the arbitrage-free derivative price.
- Illustration: R_0=6%, S_0=100, $S_1(H)$=130, $S_1(T)$=94; put K=100 pays (0,6). $\tilde{p}$=1/3, $\tilde{q}$=2/3 ⇒ $V_0 \approx$ \$3.77.
- If R_0 and the risk-neutral probabilities are given directly, arbitrage-free derivative values can be computed without separately being given a risky asset used to infer $\tilde{p}, \tilde{q}$.

The physical probability describes frequencies under the actual world. The pricing probability is chosen so that traded asset prices equal discounted expected payoffs. Its role is valuation, not forecasting; both originate from the same replication argument.

Multi-period information: adapted and predictable processes

- Trading dates n=0,…,N. $B_{n,m}$=time-n price of a \$1 ZCB maturing m≥n. R_n is the one-period spot/short rate from n to n+1, known at n but not earlier; $B_{n,n+1} = 1/(1+R_n)$.
- Negative rates do not automatically create arbitrage in the MSCF self-financing model because agents are not allowed to hold uninvested cash; if riskless cash storage is allowed, negative rates would create an arbitrage. This distinction matters in applications.

Coin-toss state space, measurability, and adapted processes

- $\Omega = \{H, T\}^N$ contains 2^N paths $\omega = (\omega_1, \ldots, \omega_N)$. A binomial product measure (BPM) with head probability a and tail probability b=1−a assigns $P(\omega) = a^{\#H} b^{\#T}$; tosses are independent.
- A random variable is time-n measurable if it depends only on the first n tosses — exactly the information known at n. A process is adapted if its time-n value is time-n measurable.

- Define $X_j = +1$ on H and -1 on T; $M_0 = 0,\ M_n = \sum_{j=1}^{n} X_j = \#H - \#T$. Under head probability 1/2 this is a symmetric random walk, a discrete analogue of Brownian motion.

Model specification and pricing

- A binomial short-rate model is characterized by an adapted R_n>-1 and a pricing/risk-neutral measure $\widetilde{\mathbb{P}}$ on Ω (usually BPM, often $\tilde{p}$=$\tilde{q}$=1/2 in the numerical illustrations).
- \$1 invested at n becomes $1 + R_n$ at n+1; borrowing and lending use the same rate.

$$D_0 = 1, \qquad D_m = \frac{1}{(1+R_0)(1+R_1)\cdots(1+R_{m-1})}$$

- D_n is the discount process; it is predictable because D_n is determined using information through time n−1.

Price at time 0 of payment Y at m:

$$\widetilde{\mathbb{E}}[D_m Y]$$

Price at time n:

$$\frac{1}{D_n}\widetilde{\mathbb{E}}_n[D_m Y]$$

Conditional valuation and the tower property

Conditional expectation properties

- Under a BPM, $\widetilde{\mathbb{E}}_n[W]$ averages over all possible future tosses using $\tilde{p}/\tilde{q}$ probabilities while holding the first n tosses fixed. $\widetilde{\mathbb{E}}_0$[W]=$\widetilde{\mathbb{E}}[W]$, $\widetilde{\mathbb{E}}_N[W]$=W.

Property	Formula
Linearity	$\widetilde{\mathbb{E}}_n[c_1X + c_2Y] = c_1\widetilde{\mathbb{E}}_n[X] + c_2\widetilde{\mathbb{E}}_n[Y]$
Measurability	$\widetilde{\mathbb{E}}_n[X]$ is time-n measurable
One-step	If X is time-$(n+1)$ measurable: $\widetilde{\mathbb{E}}_n[X] = \tilde{p}X(H\text{-child}) + \tilde{q}X(T\text{-child})$
Take out known	If X is time-n measurable: $\widetilde{\mathbb{E}}_n[XY] = X\widetilde{\mathbb{E}}_n[Y]$
Tower / iterated conditioning	If $n \le m$: $\widetilde{\mathbb{E}}_n[\widetilde{\mathbb{E}}_m[Y]] = \widetilde{\mathbb{E}}_n[Y]$

- Deterministic discount factor for integer time n is $d(n) = \widetilde{\mathbb{E}}[D_n]$. Given R process + pricing measure, the model determines the zero-coupon yield curve.
- First fundamental theorem statement used here: an arbitrage-free binomial pricing model is characterized by a probability measure under which every time-n price of a time-m payoff Y is

$$\frac{1}{D_n}\widetilde{\mathbb{E}}_n[D_m Y].$$

The measure need not be a BPM.

Write $D_0 = 1$ and $D_n = \prod_{j=0}^{n-1}(1+R_j)^{-1}$. For a claim paying V_m at m, the pricing formula is

$$\boxed{V_n = \frac{1}{D_n}\widetilde{\mathbb{E}}_n[D_m V_m]}, \qquad \boxed{V_0 = \widetilde{\mathbb{E}}[D_m V_m]}.$$

Economic intuition. A stochastic discount factor values not just *when* a payment arrives but *which rate state* accompanies that payment. The risk-neutral expectation embeds the state prices consistent with tradable hedges.

Part V — Interest-Rate Models and the Dynamics of the Curve

The curve is a cross-section of prices at one moment. A term-structure model supplies consistent rate evolution across future states and makes contingent claims computable.

Short-rate versus whole-curve frameworks

- Uncertain fixed-income cash flows require a model of interest-rate evolution.
- Short-rate models: prescribe a risk-neutral/pricing measure and short-rate dynamics under it. They are internally arbitrage-free; the spot curve is an output, and parameters are calibrated to match the observed curve. Internal consistency does not guarantee easy hedging in real market instruments.
- Whole-yield models: current spot curve is an input, then the random evolution of the whole curve is modeled. No-arbitrage restrictions are harder; Heath–Jarrow–Morton is named as the key framework and forward rates are central.
- Terminology note: short-rate models are often called equilibrium models; whole-yield models are often called arbitrage-free models.

Ho–Lee, Vasicek, Hull–White, and Black–Derman–Toy

Model	Dynamics / key feature
Ho–Lee	$\Delta R_n = \lambda_{n+1} + \sigma X_{n+1}$; $R_n = R_0 + \sum_{i=1}^{n} \lambda_i + \sigma M_n$. Can be written $a_n + b\,\#H$ with $b = 2\sigma$, or $a_n + b_n\,\#H$ for variable volatility. Recombining; flexible fit to the current term structure.
Vasicek	$\Delta R_n = k(\theta - R_n) + \sigma X_{n+1}$, $0 < k < 1$. Solution: $R_n = (1-k)^n(R_0 - \theta) + \theta + \sigma \sum_{j=1}^{n} (1-k)^{n-j} X_j$. Mean reversion; θ=long-run mean, k=speed.
Hull–White	$\Delta R_n = k(\theta_{n+1} - R_n) + \sigma_{n+1} X_{n+1}$; θ and σ can vary with time, and k may also be time-dependent.
Black–Derman–Toy	$\Delta(\ln R_n) = -(\Delta\sigma_n/\sigma_n)(\ln \theta_{n+1} - \ln R_n) + \sigma_{n+1} X_{n+1}$. Rewrites as $R_n = a_n b_n^{\#H}$, giving a recombining positive-rate tree.

Calibration versus dynamics and model quality

- Calibration chooses model parameters so model prices/term structures agree with observed market data. A central warning from the examples is: an inappropriate model cannot be rescued simply by adding many parameters until it exactly fits today's bond prices.
- Continuous-time term-structure models can be easier to calibrate because some have explicit formulas. For continuous-time Ho–Lee,

$$dr_t = \lambda_t\, dt + \sigma\, dW_t.$$

  For a fixed $\sigma > 0$, the drift process λ_t can be chosen to match the current term structure under suitable smoothness. Volatility can be estimated from historical data or from prices of caps/caplets. A continuous-time model can then be discretized to produce a binomial version.

Calibration principle. Matching observed zero prices constrains todayś cross-section, but the same price curve can be produced by distinct assumptions about volatility, mean reversion, and state dependence. Those differences can materially change option values and hedges.

Part VI — The Pricing Engine: Backward Induction and State Reduction

Risk-neutral valuation becomes an algorithm by conditioning one step at a time. This one recursion prices bonds, cash-flow streams, options, and more elaborate contracts.

The one-step pricing recursion and coupon timing

- Dates 0,...,N; $\Omega = \{H, T\}^N$; R_n adapted; pricing measure $\widetilde{\mathbb{P}}$; D_n predictable with $D_0 = 1$ and

$$D_n = \frac{1}{\prod_{j=0}^{n-1}(1 + R_j)}.$$

Single payment V_m:

$$V_n = \frac{1}{D_n}\widetilde{\mathbb{E}}_n[D_m V_m],$$

$$\text{in particular} \qquad V_0 = \widetilde{\mathbb{E}}[D_m V_m].$$

- For testing, the notes state short rates will be given as: Ho–Lee form $R_n = a_n + b_n \#H$; BDT form $R_n = a_n b_n^{\#H}$; or explicit node values. In these cases $\widetilde{\mathbb{P}}$ is a BPM with head probability 1/2.
- In abstract models, the time step is not specified; R_n always means the one-period rate. Bond face values are $1 unless stated otherwise.

ZCBs and coupon bonds

$$B_{n,m} = \frac{1}{D_n}\widetilde{\mathbb{E}}_n[D_m], \qquad B_{n,n} = 1$$

Coupon bond (face 1, one-period coupon q):

$$C^q_{n,m} = B_{n,m} + q \sum_{i=n+1}^{m} B_{n,i}$$

- For n>0, $C^q_{n,m}$ is the price just after the time-n coupon has been received.

General multiple cash flows and one-period forwards

If adapted A_n is paid at n=1,...,m:

$$V_0 = \widetilde{\mathbb{E}}\left[\sum_{n=1}^{m} D_n A_n\right]$$

$$\text{One-period forward } F_{n,m} = \frac{B_{n,m}}{B_{n,m+1}} - 1$$

- $F_{n,m}$ is fixed at n for a loan from m to m+1. At initiation time m, $F_{m,m} = R_m$.

For a no-payment security, backward induction reads

$$\boxed{V_n = \frac{\tilde{p}V^H_{n+1} + \tilde{q}V^T_{n+1}}{1 + R_n}}.$$

For a coupon or other cash flow C_{n+1} paid at the next date, include the payment *inside* the conditional expectation before discounting. Whether the value is measured just before or after the coupon is essential to avoiding double counting.

Tree valuation of bonds, swaps, and bond options

Start from the risk-neutral pricing formula for a single payment. Because $1+R_n$ is known at n:

$$V_n = \frac{1}{1+R_n}\widetilde{\mathbb{E}}_n[V_{n+1}]$$

Under a BPM:

$$V_n(\text{node}) = \frac{\tilde{p}\,V_{n+1}(\text{H child}) + \tilde{q}\,V_{n+1}(\text{T child})}{1+R_n(\text{node})}$$

- Backward induction applies a one-period model from maturity back one layer at a time. Tower/iterated conditioning guarantees equivalence to directly computing $\widetilde{\mathbb{E}}[D_m payoff]$.

Illustration — Ho–Lee tree, zeros, swap, European bond option

- Rates: $R_0 = 5\%$; $R_1(H) = 5.5\%$, $R_1(T) = 4.5\%$;
  $R_2(HH) = 6\%$, $R_2(HT) = R_2(TH) = 5\%$, $R_2(TT) = 4\%$; $\tilde{p} = \tilde{q} = 0.5$.
- Time-0 ZCB prices: $B_{0,1}$=0.9523810; $B_{0,2}$=0.9070500; $B_{0,3}$=0.8639160.
- 3-period swap/par rate: $SR_{0,3}=(1-B_{0,3})/(B_{0,1}+B_{0,2}+B_{0,3})$=4.9969%.
- European call: exercise at 2, strike \$95, underlying \$100-face ZCB maturing 3. At time 2 the bond values are 94.33962 (HH), 95.23810 (HT/TH), 96.15385 (TT), so option payoffs are 0, 0.23810, 1.15385 respectively.
- Backward values: $V_1(H)$=0.112844; $V_1(T)$=0.666005; V_0=0.370880.

Discounted-price martingales and streams of payments

- A $\widetilde{\mathbb{P}}$-martingale is an adapted process M_n with $M_n=\widetilde{\mathbb{E}}_n[M_{n+1}]$.
- For a security with one terminal payment, the discounted price process D_nV_n is a martingale under the risk-neutral measure.

Multiple cash flows

Let A_n be paid at time n and define V_n as price immediately after A_n is paid:

$$V_m = 0, \qquad V_n = \frac{1}{1+R_n}\widetilde{\mathbb{E}}_n[A_{n+1}+V_{n+1}]$$

- For multiple payments, D_nV_n alone is not a martingale because intermediate payouts leave the asset. Before/after-payment conventions change formulas, so always state the convention; final coupon and principal may be treated as one combined payment or two payments.

State compression, recombining structures, and a BDT illustration

- Naive tree enumeration has 2^N paths. Recombining trees avoid repeated work.
- If R_n depends on the path only through k=#H up to n, write $R_n = r_n(k)$. If terminal payoff also depends only on #H, then price is $V_n = v_n(k)$.

State recursion:

$$v_n(k) = \frac{\tilde{p}\,v_{n+1}(k+1) + \tilde{q}\,v_{n+1}(k)}{1+r_n(k)}$$

- If terminal payoff V_m depends on #H only through time m−1, then $V_{m-1} = V_m/(1+R_{m-1})$ first, and continue the same state recursion.

For multiple payments $a_{n+1}(k)$ depending on the preceding state and price measured after each payment:

$$v_m = 0, \qquad v_n(k) = \frac{a_{n+1}(k) + \tilde{p}\, v_{n+1}(k+1) + \tilde{q}\, v_{n+1}(k)}{1 + r_n(k)}$$

Illustration — BDT state reduction

- 5-period BDT: a_n=0.05−0.005n, b_n=1.3, $\tilde{p}$=0.5; $R_n = (0.05 - 0.005n)(1.3)^{\#H}$.
- Price a maturity-5 ZCB and caplet paying $1{,}000(R_4 - 5\%)^+$ at time 5. At time 4, $B_{4,5} = 1/(1 + R_4)$ and caplet price is payoff/(1+R_4). Backward-induct in state k=#H.
- Final time-0 results from the tables: $B_{0,5}$=0.776516; caplet value V_0=4.80061.

Computational intuition. A full N-step path tree has 2^N terminal paths; if price-relevant history can be summarized by (n, k) with $k = \#H$, only $O(N^2)$ distinct states must be valued. Such a reduction is valid only if the short rate and relevant future payoffs are functions of that compressed state.

Part VII — Interest-Rate Derivatives and Embedded Optionality

With the pricing engine in place, fixed-income derivatives are distinguished primarily by their payment formula and who controls exercise.

Floating-rate notes and the reset-to-par mechanism

Floating-rate note

- FRN maturity m: pays R_{n-1} at each n=1,…,m and face 1 at m.

$$D_n(1+R_{n-1}) = D_{n-1}$$

$$\widetilde{\mathbb{E}}[D_n R_{n-1}] = B_{0,n-1} - B_{0,n}$$

- Summing/telescoping gives FRN time-0 price =1. This is the tree-model version of "a floater resets to par." More generally, just before a coupon reset a plain floating-rate bond is worth face value.

A floating coupon fixed from the current one-period rate compensates for changing funding conditions. Immediately after reset (and under the matching discounting assumptions), the remaining floating leg tends to return to par. A capped or otherwise modified coupon need not have this property.

Swaps, forward-rate decomposition, caps and floors

Swap, cap, floor definitions

- m-period swap with fixed rate K pays $K - R_{n-1}$ at each n=1,…,m. The swap rate $SR_{0,m}$ is K making time-0 value zero.
- Cap pays $(R_{n-1}-K)^+$ each date; floor pays $(K-R_{n-1})^+$. Positive part $x^+ = x$ if $x > 0$, else 0; $x^+ - (-x)^+ = x$.

$$\text{Swap}_{0,m}^K = \widetilde{\mathbb{E}}\left[\sum_i D_i(K - R_{i-1})\right] = C_{0,m}^K - 1$$

$$\text{Cap}_{0,m}^K = \widetilde{\mathbb{E}}\left[\sum_i D_i(R_{i-1} - K)^+\right]$$

$$\text{Floor}_{0,m}^K = \widetilde{\mathbb{E}}\left[\sum_i D_i(K - R_{i-1})^+\right]$$

Floor-cap parity:

$$\text{Swap}^K + \text{Cap}^K = \text{Floor}^K$$

- A single cap payment is a caplet; a single floor payment is a floorlet.

Swap rate and forward-rate representation

$$SR_{0,m} = \frac{1 - B_{0,m}}{\sum_{i=1}^m B_{0,i}}$$

$$\text{Swap}_{0,m}^K = K\sum_{i=1}^m B_{0,i} + B_{0,m} - 1$$

$$\text{Swap}_{0,m}^K = \sum_{i=1}^m B_{0,i}[K - F_{0,i-1}]$$

- The model swap rate equals the m-period par-coupon rate. At initiation, a plain-vanilla **receiver swap** can be replicated by going long a par-coupon bond and short a floating-rate bond.
- Customized swaps can have different fixed and floating payment frequencies or changing notionals. The message is to value the actual fixed and floating cash-flow legs rather than memorize one formula.
- Market variants named in : **amortizing swaps** (scheduled declining notional), **zero swaps** (payments deferred until maturity), **putable swaps** (receiver of fixed may cancel), and **callable swaps** (payer of fixed may cancel). Swap counterparties generally post collateral.
- One may study parallel shifts of a nonflat spot curve, but different securities have different YTMs, and a spot-curve parallel shift of α changes a typical security's YTM by approximately — not exactly — α.

For a standard semiannual interest-rate swap, the floating payment at t is $(F/2)r_{t-0.5,t}$, where $r_{t-0.5,t}$ is the future six-month spot rate observed at $t-0.5$. **Time-0 PV of floating payment:**

$$(F/2)f(t)d(t)$$

- Replication: long F-face ZCB maturing at t−0.5, short F-face ZCB at t. Time-0 value $F[d(t-0.5)-d(t)] = Fd(t)f(t)/2$.

> Conceptual caution: : PV[(future realized floating rate)×cash-flow scale] can be expressed with today's forward rate by replication; this does NOT mean the realized future spot rate equals today's forward rate.

With one-period accrual, let the unit-notional fixed leg receive K each period and the floating leg pay R_{i-1}. The algebraic payoff identities

$$(R-K)^+ - (K-R)^+ = R-K, \qquad (K-R)^+ = (K-R)+(R-K)^+$$

express cap/floor parity after specifying which party pays fixed or floating. A emphreceiver-fixed swap plus a cap on floating payments is a floor on floating payments: $K-R+(R-K)^+ = (K-R)^+$. This is the convention used for the identity $Swap + Cap = Floor$.

Inverse floaters and capped/floored coupons

Inverse floaters and capped/floored coupon structures

- A plain floater's coupon rises with the reference rate; an **inverse floater**'s coupon falls as the reference rate rises. Caps and floors truncate the coupon at prescribed bounds.
- example: in a 10-period Ho–Lee tree, a capped/floored inverse floater pays $1{,}000Q_n$ at time n, where

$$Q_n = \begin{cases} 0.035, & R_{n-1} > 0.085, \\ 0.12 - R_{n-1}, & 0.035 \le R_{n-1} \le 0.085, \\ 0.085, & R_{n-1} < 0.035. \end{cases}$$

  The example is constructed so the inverse floater can be replicated by 2× a 6% fixed-coupon bond minus 1× the capped/floored floater.

European, American, and Bermudan exercise

American option algorithm

- American option: expiration m and adapted nonnegative intrinsic-value process G_n. If unexercised, holder chooses at each n whether to exercise once or continue.

$$V_m = G_m$$
$$V_n = \max\left\{G_n, \frac{1}{1+R_n}\widetilde{\mathbb{E}}_n[V_{n+1}]\right\}$$

- Optimal exercise occurs at the earliest n where $V_n = G_n$ (exercise value equals the option value).
- If R_n and G_n depend only on n and #H, use #H as the state variable.
- With positive rates, an American call on a non-dividend stock is not optimally exercised early; for a dividend-paying stock or coupon bond, early exercise may be optimal.

Bermudan option

- Bermudan: exercise allowed only on a nonempty set $E \subset \{0,\dots,N\}$; exercise at most once. Let m=max E and set $V_m = G_m$.
- At $n \notin E$: continuation value only. At $n \in E$: max(intrinsic, continuation), exactly as the American recursion.

Illustration — American call on a coupon bond

- 3-period model with $\widetilde{p}$=0.5: R_0=6%; $R_1(H)$=6.6%, $R_1(T)$=5.2%; $R_2(HH)$=8%, $R_2(HT)$=$R_2(TH)$=7%, $R_2(TT)$=5%.
- ZCBs: $B_{0,1}$=0.943396226; $B_{0,2}$=0.890875776; $B_{0,3}$=0.834670563. Par one-period coupon rate $q = (1-B_{0,3})/(B_{0,1}+B_{0,2}+B_{0,3}) = 6.1945671\%$.
- American call: expiry 2, strike \$1,000, underlying \$1,000-face par coupon bond maturing 3, exercise just after coupon payment.
- Time-2 bond prices: HH=983.2830; HT/TH=992.4726; TT=1,011.3768. Time-1: H=984.8237; T=1,011.2836.
- Intrinsic values: $G_0 = 0$; $G_1(H) = 0$, $G_1(T) = 11.283631$; at time 2 only TT is positive, $G_2(TT) = 11.376829$.
- Backward induction: $V_1(H) = 0$; $V_1(T) = \max(11.283631,$ discounted expected continuation$) = 11.283631$, so exercise early after T. $V_0 = 5.322467453$.
- If European (exercise only at time 2), time-0 price would be 2.5506. The difference is a substantial early-exercise premium.

Swaptions and exercise-date design

- A **swaption** is an OTC option to enter an interest-rate swap at a future date at a prescribed fixed rate. A receiver swaption gives the right to receive fixed; a payer swaption gives the right to pay fixed.
- Swaptions can be European, Bermudan, or American. For American/Bermudan contracts, the post-exercise swap may be specified either by a fixed final maturity date or by a fixed tenor starting when the option is exercised.
- The exercise rule is the same as for other early-exercise claims: compare immediate exercise value with discounted risk-neutral continuation value at each allowed exercise date. You should be able to analyze a new option structure, not only reproduce a memorized example.

Callable bonds, putable bonds, and negative convexity

A call owned by the issuer limits the investor's upside when falling yields make redemption attractive. A put owned by the investor protects the downside when yields rise.

- Callable bond = option-free bond − issuer's call option. If the issuer can call at par in one year,

the embedded call can be viewed as a European call on the underlying noncallable bond for the simplified example.

Underlying yield	Callable price	Callable D	Callable C
4%	100.0251	2.1557	−146.618
5%	96.9499	4.2277	−223.035
6%	91.8734	6.3709	−119.563

- The example exhibits negative convexity. At low market rates the call becomes valuable to the issuer, compressing upside in the bond price; the price–yield curve can lie below its tangent line.
- At high market rates relative to coupon, call probability is low and the callable bond behaves more like an option-free bond, generally with positive convexity.
- Yield to call = YTM computed under the assumption the bond is called at the earliest future call date.

Contract definition

Consider an N-period binomial model. A bond has maturity m, face value F, one-period coupon rate q, exercise dates $E \subset \{1, \dots, m-1\}$, and call/put prices $(F_n)_{n\in E}$.

- **Callable bond:** at $n \in E$, if the bond is still alive, the issuer may pay $Fq + F_n$ and terminate all future obligations.
- **Putable bond:** at $n \in E$, if the bond is still alive, the holder may sell it back after the coupon payment and receive $Fq + F_n$, with no later payments.
- Thus exercise produces the same cash amount $Fq + F_n$; the difference is *who owns the exercise right*: issuer for a call, holder for a put.

Option identity and one-pass backward induction

The useful algebra is

$$x - (x-K)^+ = \min\{x, K\}, \qquad x + (K-x)^+ = \max\{x, K\}.$$

This is why a callable bond can be viewed as straight bond minus a call option, and a putable bond as straight bond plus a put option.

Let V_n^c and V_n^p denote the post-coupon prices of the callable and putable bonds. Set the terminal post-coupon value to the principal, $V_m^c = V_m^p = F$. If $n \notin E$,

$$V_n^c = \frac{1}{1+R_n}\widetilde{\mathbb{E}}_n[V_{n+1}^c + Fq], \qquad V_n^p = \frac{1}{1+R_n}\widetilde{\mathbb{E}}_n[V_{n+1}^p + Fq].$$

If $n \in E$,

$$\boxed{V_n^c = \min\left\{F_n, \frac{1}{1+R_n}\widetilde{\mathbb{E}}_n[V_{n+1}^c + Fq]\right\}}$$

and

$$\boxed{V_n^p = \max\left\{F_n, \frac{1}{1+R_n}\widetilde{\mathbb{E}}_n[V_{n+1}^p + Fq]\right\}}.$$

The continuation value is compared with the contractual exercise price *after the current coupon has been paid*.

Callable: issuer chooses the cheaper liability ⇒ **min**. Putable: holder chooses the more valuable alternative ⇒ **max**.

Part VIII — Forwards, Futures, and the Economics of Settlement

A forward and a futures contract can reference the same terminal asset but settle their gains on different dates. That timing interacts with stochastic financing rates.

Forward contracts: delivery payment and cash-and-carry

Forward contracts

- Forward: bilateral agreement to trade an asset at future delivery date m for fixed delivery price K. Short sells, long buys. K is set so both positions have zero value when entered.
- $\text{For}_{n,m}$ = the delivery price at time n that makes a newly initiated forward to m worth zero. As time passes, an existing forward generally has nonzero long/short values, equal and opposite.

If the underlying security's time-m price is P_m:

$$\text{For}_{n,m} = \frac{\widetilde{\mathbb{E}}_n[D_m P_m]}{\widetilde{\mathbb{E}}_n[D_m]}$$

- Do not assume $\widetilde{\mathbb{E}}_n[D_m P_m] = D_n P_n$ in general. If the security pays no dividends/coupons from n to m, then it does hold, and:

$$\text{For}_{n,m} = \frac{P_n}{B_{n,m}}$$

Definitions and pricing formulas

- Work in the binomial framework with an underlying security whose price at delivery date m is $P_m(\omega_1, \dots, \omega_m)$.
- A **long forward** initiated at time $n \le m$ has no cash flow before m. Its time-m payoff is

$$P_m - \text{For}_{n,m}.$$

- A **long futures** position initiated at time $j \le m$ costs zero to enter and is marked to market each period. At time $k+1$ it receives

$$\text{Fut}_{k+1,m} - \text{Fut}_{k,m}, \qquad k = j, \dots, m-1,$$

  with $\text{Fut}_{m,m} = P_m$. It can be closed at no cost at any time.

The two pricing formulas are

$$\boxed{\text{For}_{n,m} = \frac{\widetilde{\mathbb{E}}_n[D_m P_m]}{\widetilde{\mathbb{E}}_n[D_m]}} \quad \text{and} \quad \boxed{\text{Fut}_{n,m} = \widetilde{\mathbb{E}}_n[P_m]}.$$

If the underlying security pays no dividends or coupons between n and m, then

$$\widetilde{\mathbb{E}}_n[D_m P_m] = D_n P_n, \qquad \widetilde{\mathbb{E}}_n[D_m] = D_n B_{n,m},$$

so

$$\boxed{\text{For}_{n,m} = \frac{P_n}{B_{n,m}}}.$$

Futures contracts: marking to market and pricing martingales

Futures mechanics

- Forwards face counterparty matching/default problems. Futures use a central exchange and margin accounts; all contracts for the same asset/delivery date use the same futures delivery price.

- Futures are marked to market daily. If the futures price rises, longs receive cash and shorts pay cash; the reverse occurs when it falls. Low margin balances trigger margin calls. Because gains/losses are settled through the margin account, closing early leaves no further net payment owed on the futures position.

Futures-price process and proposition

An m-futures process $\mathrm{Fut}_{n,m}$ is adapted and satisfies:

$$(i)\mathrm{Fut}_{m,m} = P_m$$

$$(ii)\widetilde{\mathbb{E}}_n[\sum_{i=n}^{m-1} D_{i+1}(\mathrm{Fut}_{i+1,m} - \mathrm{Fut}_{i,m})] = 0$$

- A long receives $\mathrm{Fut}_{k+1,m} - \mathrm{Fut}_{k,m}$ on day k+1. Condition (ii) says the time-n value of all future margin adjustments is zero.

Unique futures process:

$$\mathrm{Fut}_{n,m} = \widetilde{\mathbb{E}}_n[P_m]$$

Derivation:

1. Set $\mathrm{Fut}_{n,m}=\widetilde{\mathbb{E}}_n[P_m]$. At m, time-m measurability gives $\mathrm{Fut}_{m,m}=P_m$. 2. For each future daily increment, use tower property and "take out known" for D_{i+1} to show its time-n expected discounted value is zero; summing proves condition (ii). 3. For uniqueness, subtract condition (ii) starting at k+1 from condition (ii) starting at k; after conditioning at k, obtain $\widetilde{\mathbb{E}}_k[D_{k+1}(\mathrm{Fut}_{k+1,m} - \mathrm{Fut}_{k,m})]=0$. Since D_{k+1} is known and nonzero at k, $\mathrm{Fut}_{k,m}=\widetilde{\mathbb{E}}_k[\mathrm{Fut}_{k+1,m}]$. Recursing to m yields $\widetilde{\mathbb{E}}_k[P_m]$.

Forward vs futures: Forward price uses a discount-weighted conditional expectation, while futures price is the ordinary risk-neutral conditional expectation of the terminal asset price. With stochastic interest rates, the two need not be equal.

Forward–futures basis and stochastic-rate covariance

Forward price versus futures price

At time 0 the sign of the relation is controlled by the risk-neutral correlation between the discount factor D_m and terminal security price P_m:

- if D_m and P_m are uncorrelated, $\mathrm{For}_{0,m} = \mathrm{Fut}_{0,m}$;
- if they are positively correlated, $\mathrm{For}_{0,m} > \mathrm{Fut}_{0,m}$;
- if they are negatively correlated, $\mathrm{For}_{0,m} < \mathrm{Fut}_{0,m}$.

The same idea applies conditionally for $n > 0$.

Conceptual caution: A forward and a futures contract can have the same terminal economic exposure but different prices when rates are stochastic, because futures gains/losses are realized earlier and can be reinvested at state-dependent rates.

At time zero, directly subtract the two pricing expressions:

$$\boxed{\mathrm{For}_{0,m} - \mathrm{Fut}_{0,m} = \frac{\mathrm{Cov}_{\widetilde{\mathbb{P}}}(D_m, P_m)}{\widetilde{\mathbb{E}}[D_m]}.}$$

The covariance is under the pricing measure. Deterministic discounting makes it zero, but stochastic rates permit positive or negative deviations. This is a settlement-timing effect rather than a difference in the terminal quoted underlying.

Coupon-bond delivery, financing payments, and Ho–Lee comparisons

Illustration — delivery of a coupon bond

Use the three-period tree in the American coupon-bond call example and deliver at time 2 a par coupon bond with face value 1,000 and maturity 3. Delivery is **ex-coupon**; W_n denotes the bond price just after coupon payment. At time 2,

$$W_2(HH) = 983.283, \quad W_2(HT) = W_2(TH) = 992.473, \quad W_2(TT) = 1{,}011.377.$$

The required expectations are

$$\widetilde{\mathbb{E}}[D_2] = 0.890876, \qquad \widetilde{\mathbb{E}}[D_2 W_2] = 886.3747.$$

Hence

$$\mathrm{For}_{0,2} = 994.948, \qquad \mathrm{Fut}_{0,2} = \widetilde{\mathbb{E}}[W_2] = 994.901.$$

Replication check. Buying the bond today gives coupons before delivery that a forward holder does not receive, so these coupons must be stripped out. The coupon is 61.945671 at times 1 and 2, giving zero initial cost condition

$$1{,}000 - \mathrm{For}_{0,2} B_{0,2} - 61.945671(B_{0,1} + B_{0,2}) = 0,$$

which gives

$$\mathrm{For}_{0,2} = 994.94768.$$

Illustration — an interest-rate payoff and cash-flow timing

Consider a security that pays $10{,}000R_2$ at time 3. Since

$$\widetilde{\mathbb{E}}[D_3 R_2] = \widetilde{\mathbb{E}}[D_3(1 + R_2)] - \widetilde{\mathbb{E}}[D_3] = B_{0,2} - B_{0,3},$$

we get

$$\frac{\mathrm{For}_{0,3}}{10{,}000} = \frac{B_{0,2}}{B_{0,3}} - 1 = F_{0,2}.$$

Numerically,

$$\mathrm{For}_{0,3} = 673.382, \qquad \mathrm{Fut}_{0,3} = 675.$$

The futures-price process is

$$\mathrm{Fut}_{1,3}(H) = 750, \quad \mathrm{Fut}_{1,3}(T) = 600,$$
$$\mathrm{Fut}_{2,3}(HH) = 800, \quad \mathrm{Fut}_{2,3}(HT) = \mathrm{Fut}_{2,3}(TH) = 700, \quad \mathrm{Fut}_{2,3}(TT) = 500.$$

State class	Forward cash flow at $t = 3$	Futures cash flow at $t = 1$	Futures cash flow at $t = 2$
$HH*$	126.618	+75	+50
$HT*$	26.618	+75	−50
$TH*$	26.618	−75	+100
$TT*$	−173.382	−75	−100

For this security the time-3 futures cash flow is zero because $\mathrm{Fut}_{3,3} = \mathrm{Fut}_{2,3}$: the payoff $10{,}000R_2$ is already known at time 2. The forward still has only one cash flow at time 3. If the earlier futures cash flows are invested in the bank account until time 3, their total time-3 values are

$$140.36\ (HH*), \quad 32.0465\ (HT*), \quad 22.577\ (TH*), \quad -187.845\ (TT*).$$

This illustrates why forward and futures positions need not generate the same terminal wealth under stochastic rates.

The **value** of the existing long forward changes even though no cash is exchanged before maturity. In this example it is +66.658 after H and −66.658 after T at time 1; at time 2 it is 117.239 at HH, 24.877 at HT or TH, and −165.126 at TT.

Illustration — Ho–Lee zero-coupon bond

In the stated three-period Ho–Lee model, for a face-100 zero-coupon bond maturing at time 3 and delivered at time 2,

$$\mathrm{For}_{0,2} = 95.24458, \qquad \mathrm{Fut}_{0,2} = 95.24242.$$

At time 1 the two prices coincide state by state:

$$\mathrm{For}_{1,2}(H) = \mathrm{Fut}_{1,2}(H) = 94.78886,$$

$$\mathrm{For}_{1,2}(T) = \mathrm{Fut}_{1,2}(T) = 95.69598.$$

Part IX — Measuring Interest-Rate Risk

A pricing function specifies current value. Its derivatives ask how value changes when the chosen interest-rate coordinate moves.

From Taylor expansion to duration and DV01

- Even default-free deterministic-cash-flow securities have interest-rate risk: as rates rise, previously issued fixed-rate bond prices generally fall.
- Yield-curve moves can be complex; short and long rates can even move in opposite directions. Practical risk measurement compresses the curve into one or a few key factors. A parallel-shift factor is a special case.
- One-factor model: P=f(y), where y is one chosen rate factor. Very simple, but not always appropriate.

First order:

$$\Delta P \approx f'(y)\Delta y$$

Second order:

$$\Delta P \approx f'(y)\Delta y + \tfrac{1}{2} f''(y)(\Delta y)^2$$

ZCB sensitivity and basis points

ZCB:

$$P = F\left(1 + \frac{y}{2}\right)^{-2T}$$
$$\frac{dP}{dy} = -\frac{T}{1 + y/2}\,P$$

First-order:

$$\Delta P \approx -\frac{T}{1 + y/2}\,P\,\Delta y$$

- A 1 bp change is 0.01 percentage point = 0.0001 in decimal rate units. Example: 4.853% + 2 bp = 4.873%.

DV01, hedging, duration, and convexity

$$\text{DV01} = -\frac{\Delta P}{10{,}000\,\Delta y} \approx -\frac{f'(y)}{10{,}000}$$

- DV01 = dollar value of an 01; PVBP means the same thing. Here the sign convention makes DV01 positive for standard fixed-cash-flow bonds because price falls as yield rises. Market sign conventions can differ; always specify units/face amount.
- Market participants usually mean yield-based DV01: y is the security's YTM. Other factors are possible, so clarify the variable.
- Technical note: for a *random* payoff that is guaranteed to be positive, DV01 need not be positive. Positivity of DV01 is a property of standard deterministic positive-cash-flow bonds, not of every positive random claim.
- Call-option example: option price 8.0866 at $y = 4.01\%$ and 8.2148 at 3.99% ⇒ centered DV01 $= 0.0641$ per $100 face.

A +1 bp yield change is expected to reduce option value by about 6.41 cents per $100 face.

- Hedge example: short $100mm face of the call; underlying bond DV01=0.0857 per $100. Match DV01 by buying $100mm×0.0641/0.0857=$74,795,799.30 face of the bond.
- Sign rule: if A and B have same-sign DV01, hedge a short A with long B. If DV01 signs are opposite, hedge a short A with short B.

$$\text{Duration } D = -\frac{1}{P}\frac{\Delta P}{\Delta y} \approx -\frac{f'(y)}{P}$$

$$\text{DV01} = \frac{PD}{10{,}000}$$

- Duration measures relative price sensitivity; DV01 measures dollar sensitivity. Enter Δy in decimals: 1 bp=0.0001, 1 percentage point=0.01.

$$\text{Convexity } C = \frac{f''(y)}{P} = \frac{1}{P}\frac{d^2P}{dy^2}$$

- If no analytical price formula exists, approximate convexity numerically.

$$P = f(y)$$

$$\text{DV01} = -\frac{f'(y)}{10{,}000},$$

$$D = -\frac{f'(y)}{P},$$

$$C = \frac{f''(y)}{P}.$$

Difference quotients:

$$\text{DV01} \approx -\frac{\Delta P}{10{,}000\,\Delta y}, \qquad D \approx -\frac{\Delta P}{P\,\Delta y}$$

$$\frac{\Delta P}{P} \approx -D\,\Delta y$$

$$\frac{\Delta P}{P} \approx -D\,\Delta y + \frac{1}{2}C(\Delta y)^2$$

- Some authors define convexity as $\frac{f''(y)}{2P}$, in which case the explicit 1/2 disappears from the second-order approximation.

Sign	Price implication
$D > 0$	Rates fall ⇒ price rises; rates rise ⇒ price falls.
$D < 0$	Rates fall ⇒ price falls; rates rise ⇒ price rises.
$C > 0$	Convexity term is positive for either sign of Δy; “long volatility.”
$C < 0$	Convexity term is negative for either sign of Δy; “short volatility.”

Finite differences and non-smooth option values

Numerical differentiation and step-size control

When a closed-form price function is unavailable, reprice the security after a small change in the chosen rate factor. For a price function $f(y)$:

$$f'(y) \approx \frac{f(y+h) - f(y)}{h}, \qquad f'(y) \approx \frac{f(y+h) - f(y-h)}{2h},$$

$$f''(y) \approx \frac{f(y+h) - 2f(y) + f(y-h)}{h^2}.$$

- Centered differences usually improve the first-derivative approximation. Always test more than one step size: a step that is too large creates truncation error; a step that is too small can create numerical noise.
- Plain-vanilla bond convexity is usually numerically stable. Exotic securities can have price functions with kinks/non-smooth points because exercise rules, caps/floors, or state changes switch discretely. Around such points, estimated DV01 and especially convexity can be sensitive to h.

Macaulay duration and cash-flow-weighted sensitivity

For a coupon bond with semiannual coupons:

$$P = F\left[\left(1+\frac{y}{2}\right)^{-2T} + \frac{q}{2}\sum_{i=1}^{2T}\left(1+\frac{y}{2}\right)^{-i}\right]$$

$$\text{DV01} = \frac{F}{10{,}000(1+y/2)}\left[\frac{T}{(1+y/2)^{2T}} + \frac{q}{2}\sum_{i=1}^{2T}\frac{i/2}{(1+y/2)^{i}}\right]$$

Equivalent compact DV01:

$$\text{DV01} = \frac{F}{10{,}000}\left[\frac{q}{y^2}\left(1-(1+y/2)^{-2T}\right) + \left(1-\frac{q}{y}\right)\frac{T}{(1+y/2)^{2T+1}}\right]$$

Macaulay duration

For nonnegative payments F_i at $T_i = i/2$:

$$D_{\text{Mac}} = \frac{1}{P}\sum_i T_i \frac{F_i}{(1+y/2)^{i}}$$

- Interpretation: PV-weighted average payment time, where discounting uses the security's YTM.

$$D_{\text{Mac}} = \left(1+\frac{y}{2}\right)D$$

- For a ZCB, $D_{\text{Mac}} = T$ exactly. D_{Mac} and D are numerically close at ordinary yields.

First-order via Macaulay:

$$\frac{\Delta P}{P} \approx -\frac{D_{\text{Mac}}}{1+y/2}\,\Delta y$$

Coupon-bond duration formulas

$$D = \frac{F}{P(1+y/2)}\left[\frac{T}{(1+y/2)^{2T}} + \frac{q}{2}\sum_i \frac{i/2}{(1+y/2)^{i}}\right]$$

$$D_{\text{Mac}} = \frac{F}{P}\left[\frac{T}{(1+y/2)^{2T}} + \frac{q}{2}\sum_i \frac{i/2}{(1+y/2)^{i}}\right]$$

- Illustration: q=4%, T=20, y=4.231% ⇒ P=96.90351 per 100; D=13.555; $D_{\text{Mac}} = 13.842$.
- Illustration: F=\$100,000 for the same bond ⇒ DV01=\$131.353. A −17 bp move predicts +\$2,233; exact repricing gives +\$2,266.59 (new price \$99,170.10). First-order error reflects curvature.

Special cases: zeros, par bonds, annuities, perpetuities

Par coupon bonds

$$D = \frac{1}{y}\left[1 - \left(1 + \frac{y}{2}\right)^{-2T}\right]$$

$$D_{\text{Mac}} = \left(\frac{1}{y} + \frac{1}{2}\right)\left[1 - \left(1 + \frac{y}{2}\right)^{-2T}\right]$$

$$\text{DV01} = \frac{F}{10{,}000\, y}\left[1 - \left(1 + \frac{y}{2}\right)^{-2T}\right]$$

- Illustration: a 10-year par yield around 4.77% gives D≈7.880, i.e. DV01≈$7.88 per $10,000 face. using $D \approx 8$ as a rule-of-thumb illustration: +4 bp on $10,000 face ≈ −$32; on $1mm face ≈ −$3,200.
- Historical rate observations demonstrate the rule “lower yield ⇒ higher duration” for a given par maturity: e.g. 10-year D≈9.66 at 0.67% (Sep-2020) vs ≈7.435 at 6.01% (Jul-2000).
- A real on-the-run bond need not be exactly par or exactly 10 years to maturity. a representative example with a 4.25% coupon and 4.07% yield gives D≈8.10 and DV01≈$8.22/$10,000 face; DV01 can be larger even while duration is smaller because price is larger.

Annuities and perpetuities

Semiannual annuity:

$$D = \frac{1}{y} - \frac{1}{1 + y/2} \cdot \frac{T}{(1 + y/2)^{2T} - 1}$$

$$D_{\text{Mac}} = \frac{1}{y} + \frac{1}{2} - \frac{T}{(1 + y/2)^{2T} - 1}$$

Perpetuity:

$$P = \frac{2A}{y}, \qquad D = \frac{1}{y}, \qquad D_{\text{Mac}} = \frac{1}{y} + \frac{1}{2}$$

Convexity and the standard bond-risk relationships

- For deterministic cash flows, “convexity” under the present conventions means second derivative of price with respect to YTM divided by price.

$$\text{Coupon-bond } C = \frac{F}{P(1 + y/2)^2}\left[\frac{T(T + 0.5)}{(1 + y/2)^{2T}} + \frac{q}{2}\sum_i \frac{i(i+1)/4}{(1 + y/2)^i}\right]$$

ZCB sensitivity formulas and maturity effects

$$\text{ZCB DV01} = \frac{FT}{10{,}000(1 + y/2)^{2T+1}}$$

$$D = \frac{T}{1 + y/2}, \qquad D_{\text{Mac}} = T, \qquad C = \frac{T(T + 0.5)}{(1 + y/2)^2}$$

- For fixed y>0, ZCB DV01 increases with T until T*=1/[2 ln(1+y/2)]≈1/y, then decreases and tends to 0 as T→∞. This can occur in actual markets.
- For fixed y, ZCB convexity grows quadratically with maturity. For fixed T, it decreases as y rises.

Coupon-bond limiting values and rules of thumb

For fixed q,y>0:

$$D_{\text{Mac}} \to \frac{1}{2} + \frac{1}{y} \qquad \text{as } T \to \infty$$

Perpetuity:

$$\text{DV01} = \frac{2A}{10{,}000y^2}, \qquad D = \frac{1}{y}, \qquad D_{\text{Mac}} = \frac{1}{2} + \frac{1}{y}, \qquad C = \frac{2}{y^2}$$

Measure	Typical relationship	Important qualification
Duration	↑ maturity; ↓ coupon; ↓ yield	If $q < y$, duration eventually peaks at a very large critical maturity T^* then declines. If $q \geq y$ it increases with maturity. Example T^*: ∼ 53y for $y = 5\%, q = 1\%$; ∼ 27y for $y = 10\%, q = 2\%$.
DV01	↑ maturity; ↑ coupon; ↓ yield	For $q < y$, DV01 can peak at T^{**} and then decline. Limit as $T \to \infty$: $Fq/(10{,}000y^2)$.
Convexity	↑ maturity; ↓ coupon; ↓ yield	Maturity rule can fail for deeply discounted bonds.

Deriving the nine rules from PV weights

For deterministic nonnegative cash flows under this treatment's semiannual-YTM convention, define

$$a = 1 + \frac{y}{2}, \qquad t_i = \frac{i}{2}, \qquad w_i = \frac{CF_i a^{-i}}{P}, \qquad \sum_i w_i = 1.$$

Then

$$\boxed{D_{\text{Mac}} = \sum_i w_i t_i}, \qquad \boxed{D = \frac{1}{a}\sum_i w_i t_i},$$

and

$$\boxed{C = \frac{1}{a^2}\sum_i w_i\, t_i(t_i + 0.5)}, \qquad \boxed{\text{DV01} = \frac{PD}{10{,}000}}.$$

Thus duration is essentially a PV-weighted first moment of payment time, while convexity is essentially a PV-weighted second-moment measure. These representations make the usual rules of thumb transparent.

Measure	Rule	Mathematical reason	Intuition
Duration	$T \uparrow \Rightarrow D \uparrow$ typically	Increasing maturity normally places more PV weight at larger t_i, increasing $\sum_i w_i t_i$. For deep-discount bonds this can eventually fail; see the maturity limits below.	Cash flows arrive later, so price is usually more rate-sensitive.
Duration	$q \uparrow \Rightarrow D \downarrow$	At fixed T, y, a higher coupon raises the PV weight of the coupon stream relative to principal. The coupon stream has a PV-weighted mean payment time strictly below T, so $D_{\text{Mac}} \downarrow$; since a is fixed, $D = D_{\text{Mac}}/a \downarrow$.	Higher coupon returns more value earlier.

Duration	$y \uparrow \Rightarrow D \downarrow$	As y rises, a^{-i} penalizes long-dated cash flows more strongly. In fact, $\frac{d}{dy}\sum_i w_i t_i = -\mathrm{Var}_w(t)/a < 0$, and $1/a$ also falls.	Higher yield shifts PV weight toward earlier payments.
DV01	$T \uparrow \Rightarrow$ DV01 $\uparrow$ typically	DV01 $= PD/10{,}000$. Longer maturity usually raises D, which normally dominates; for sufficiently discounted bonds, the fall in P can eventually dominate.	Longer bonds usually carry more dollar rate exposure.
DV01	$q \uparrow \Rightarrow$ DV01 $\uparrow$	For a coupon bond, DV01 $= \frac{F}{10{,}000a}\left[Ta^{-2T} + \frac{q}{2}\sum_i t_i a^{-i}\right]$, hence $\partial\text{DV01}/\partial q > 0$.	A higher coupon lowers relative duration but increases dollar cash flows and bond value enough to raise dollar sensitivity.
DV01	$y \uparrow \Rightarrow$ DV01 $\downarrow$	DV01 $= -P'(y)/10{,}000$, so $\partial\text{DV01}/\partial y = -P''(y)/10{,}000 = -PC/10{,}000 < 0$ for ordinary positive-cash-flow bonds.	Higher yield reduces both price and dollar rate sensitivity.
Convexity	$T \uparrow \Rightarrow C \uparrow$ typically	$C = a^{-2}\sum_i w_i g(t_i)$, where $g(t) = t(t+0.5)$ increases roughly like t^2. Moving value to later dates therefore usually raises convexity.	Convexity puts especially large weight on distant cash flows.
Convexity	$q \uparrow \Rightarrow C \downarrow$	At fixed T, y, higher coupon shifts PV weight from terminal principal at T toward the coupon stream. Since the coupon stream's weighted average of $g(t)$ is below $g(T)$, convexity falls.	Higher coupons bring value forward and reduce long-end curvature exposure.
Convexity	$y \uparrow \Rightarrow C \downarrow$	Both effects are negative: $a^{-2} \downarrow$, and the PV weights shift earlier. More formally, $d\,\mathbb{E}_w[g(t)]/dy = -\mathrm{Cov}_w(g(t), t)/a < 0$ because $g(t)$ is increasing.	Higher yield makes distant cash flows less important.

Important maturity qualifications

The three "increases with maturity" rules are typical, not unconditional. For fixed $q, y > 0$, an extremely long coupon bond converges to a coupon perpetuity because the PV of terminal principal vanishes. Hence

$$D \to \frac{1}{y}, \qquad \text{DV01} \to \frac{Fq}{10{,}000y^2}, \qquad C \to \frac{2}{y^2} \qquad (T \to \infty).$$

Therefore, when $q < y$, duration can peak at a large T^* and then decline; DV01 can similarly peak and decline. Convexity's maturity rule can also fail for deeply discounted bonds.

Exact proof of the coupon-duration rule

Let

$$A = Fa^{-N}, \qquad B = \frac{F}{2}\sum_{i=1}^{N} a^{-i}, \qquad M = \frac{F}{2}\sum_{i=1}^{N} \frac{i}{2}a^{-i}, \qquad N = 2T.$$

Then

$$P = A + qB, \qquad D_{\text{Mac}}(q) = \frac{TA + qM}{A + qB}.$$

Holding T and y fixed,

$$\frac{\partial D_{\text{Mac}}}{\partial q} = \frac{A(M - BT)}{(A + qB)^2}.$$

But

$$\frac{M}{B} = \frac{\sum_{i=1}^{N}(i/2)a^{-i}}{\sum_{i=1}^{N} a^{-i}} < T,$$

because it is the PV-weighted average payment time of the coupon stream, whose payments occur from time 0.5 through T and include payments strictly before T. Thus $M < BT$ and

$$\boxed{\frac{\partial D_{\text{Mac}}}{\partial q} < 0}.$$

Economically, a higher coupon shifts value from the terminal principal toward earlier coupon payments. Since $D = D_{\text{Mac}}/(1 + y/2)$ at fixed y, modified duration falls as well.

Three-line memory aid

$$\boxed{D \sim \text{PV-weighted } t}, \qquad \boxed{C \sim \text{PV-weighted } t^2}, \qquad \boxed{\text{DV01} = \frac{PD}{10{,}000}}.$$

Hence, typically: maturity ↑ shifts cash flows later so D, C ↑; coupon ↑ shifts value earlier so D, C ↓ but DV01 ↑; yield ↑ shifts PV weight earlier so $D, \text{DV01}, C$ ↓.

Under semiannual YTM discounting, put $a = 1 + y/2$, $t_i = i/2$, and $w_i = CF_i a^{-i}/P$. Then

$$\boxed{D_{\text{Mac}} = \sum_i w_i t_i, \qquad D = \frac{1}{a}\sum_i w_i t_i, \qquad C = \frac{1}{a^2}\sum_i w_i t_i(t_i + 0.5).}$$

Economic intuition. Macaulay duration is an average time weighted by present-value contribution; modified duration scales that time for the compounding convention. Convexity is a second-moment-like measure that emphasizes more distant payments even more strongly.

Rate sensitivities of model-priced options and claims

Short-rate DV01 from the first tree step

If a security has time-1 prices $V_1(H)$ and $V_1(T)$, define the short-rate DV01 approximation by

$$\text{DV01}_{SR} = -\frac{V_1(H) - V_1(T)}{10{,}000\,[R_1(H) - R_1(T)]}.$$

- This estimates sensitivity to the short-rate factor using the up/down states. Two cautions: the rate gap $R_1(H) - R_1(T)$ may be too large for a precise local derivative, and the estimate is naturally a time-1 rather than exact time-0 DV01. With small time steps these issues are less important; with annual steps it may only be a ballpark number.

Spreadsheet bump examples and intuition

To estimate rate risk, bump the initial Ho–Lee rate R_0 and use the finite-difference formulas given above.

Security	Duration estimate	Key lesson
6% fixed-coupon bond	7.33 one-sided with 10 bp; 7.37 centered; 7.369 with 1 bp	Exact comparable formula gives $D = 7.360$, so the bump approximation is stable and accurate.
Capped/floored floater	0.1813 with 10 bp; 0.2598 with 1 bp	A plain uncapped/unfloored floater would have duration 0 at reset; cap/floor optionality creates small but nonzero sensitivity, and the estimate is more step-size sensitive.
Capped/floored inverse floater	14.45 with 10 bp	Much larger duration, about twice the fixed bond in this example. Replication $P^{(3)} = 2P^{(1)} - P^{(2)}$ gives the same price and duration as a consistency check.

Part X — Portfolio Risk and the Parallel-Shift Approximation

Dollar exposures add across holdings. Relative exposures are meaningful only after specifying the capital base and the yield-curve perturbation.

Portfolio DV01, duration, convexity, and position signs

- DV01 is a dollar exposure, so it scales with position size. Duration and convexity are relative exposures, so multiplying the entire position does not change them.

For holdings α_i in securities with dollar prices P_i and total net price $X = \sum_i \alpha_i P_i$ (α_i may be negative):

$$\mathrm{DV01}_{\mathrm{port}} = \sum_i \alpha_i \, \mathrm{DV01}_i$$

$$D_{\mathrm{port}} = \sum_i \left(\frac{\alpha_i P_i}{X} \right) D_i$$

$$C_{\mathrm{port}} = \sum_i \left(\frac{\alpha_i P_i}{X} \right) C_i$$

- Duration/convexity weights are percentages of net capital. under the present conventions, portfolio D and C terminology is used only when net price X>0. DV01 remains meaningful for zero/negative net-price portfolios.
- If two portfolios have the same price and same duration, they have the same DV01 and approximately the same small-yield price change. Matching price + duration + convexity improves a hedge.
- Given any target P>0, duration δ, convexity γ and three distinct maturities $T_1<T_2<T_3$, one can construct a 3-ZCB portfolio matching all three quantities.
- **Dollar duration and dollar convexity**: $D_{\$} = -dP/dy = PD$ and $C_{\$} = d^2P/dy^2 = PC$. Dollar duration is $10{,}000 \times$ DV01. Dollar convexities add directly across positions, which makes them especially convenient for portfolio hedging.

An explicit portfolio aggregation illustration

Portfolio example — Illustration

- Flat spot curve 8%. Portfolio A: \$500k market value in a 10-year par bond + \$1mm market value in a 20-year ZCB.
- 10-year par $D_{\mathrm{Mac}} = 7.06697$; ZCB $D_{\mathrm{Mac}} = 20$. Portfolio $D_{\mathrm{Mac}} = [500k \times 7.06697 + 1mm \times 20]/1.5mm = 15.689$.
- $\mathrm{DV01}_A$ = (1.5mm/10,000)×(15.689/1.04) = \$2,262.84.
- A single 15-year par bond can first-order hedge A by matching DV01. Required face ≈ \$2,617,200.
- A parallel −24 bp shift predicts $\Delta P_A \approx +\$2{,}262.84 \times 24 = +\$54{,}308$.

For a common small change in the yield coordinate y, a portfolio with total market value $P_{\mathrm{port}} > 0$ has

$$\Delta P_{\mathrm{port}} \approx -P_{\mathrm{port}} D_{\mathrm{port}} \Delta y + \tfrac{1}{2} P_{\mathrm{port}} C_{\mathrm{port}} (\Delta y)^2.$$

When individual parts react to different rate maturities, a common Δy is an assumption, not an empirical law. This motivates key-rate and multifactor descriptions.

Part XI — Key-Rate and Multifactor Yield-Curve Risk

Real yield curves move in level, slope, and curvature, not just in parallel. A risk measure for one rate is therefore replaced by a vector of exposures to selected curve coordinates.

Why a single curve factor is inadequate

- One-factor hedging implicitly treats all maturities as perfectly correlated. A 6-month rate cannot realistically be expected to perfectly predict a 30-year rate.
- Complete immunization matches every liability cash flow with a ZCB. It requires no model of curve movement but is costly, often impractical, and sometimes impossible.
- Three or four rate factors often capture substantially more curve risk than a one-factor model while remaining practical.

Key-rate perturbations, sensitivities, and discount-factor reconstruction

- Choose what rate to shock (spot, par-coupon yield, etc.), how many key rates, and maturities. Spot rates are natural when hedging with zeros; par yields are natural when hedging with par bonds.
- The key-rate illustration uses 2-, 5-, 10-, and 30-year par-coupon yields $y_1,\ldots,y_4$. A shock to a key rate should leave the other key rates unchanged but influence nearby maturities more strongly than distant maturities.

Key-rate DV01, duration, and convexity

$$P = f(y_1, y_2, y_3, y_4)$$

$$\Delta P \approx \sum_i \frac{\partial f}{\partial y_i}\,\Delta y_i$$

$$\text{Key-rate DV01}_i = -\frac{1}{10{,}000}\frac{\partial f}{\partial y_i}$$

$$\text{Key-rate duration}_i = -\frac{1}{P}\frac{\partial f}{\partial y_i}$$

Second order:

$$\Delta P \approx \sum_i f_i \Delta y_i + \frac{1}{2}\sum_{i,j} f_{ij}\Delta y_i \Delta y_j$$

- With multiple factors, duration is a vector and convexity is a matrix (the Hessian scaled by price).

Shock functions and perturbed curve

Each 1 bp key-rate change generates a piecewise-linear influence function $Y_i(t)$, equal to 1 at its own key maturity and 0 at the other key maturities. Influence is largest nearby.

Function	Guide schematic
$Y_1(t)$, 2y key	≈1 at short maturities through 2y; linearly falls to 0 by 5y; 0 thereafter.
$Y_2(t)$, 5y key	0 near 2y; triangular peak 1 at 5y; back to 0 at 10y.
$Y_3(t)$, 10y key	0 at 5y; peak 1 at 10y; linearly declines to 0 at 30y.
$Y_4(t)$, 30y key	0 through 10y; linearly rises to 1 at 30y.

Perturbed curve:

$$y(t) = y(t) + \sum_{i=1}^{4}(y_i - y_i^*)Y_i(t)$$

- The resulting curve is an approximation to the true new curve, but with several key rates it can be a good one. The price is treated as $f(y_1, \dots, y_4)$, with the reference price at $y_i = y_i^*$.
- In practice, compute partial derivatives through numerical repricing/difference quotients rather than derive a complicated explicit price formula.

Par-coupon yields → discount factors → price

Par-coupon yield at maturity (n+1)/2 satisfies:

$$y\left(\frac{n+1}{2}\right) = \frac{2\left[1 - d\left(\frac{n+1}{2}\right)\right]}{\sum_{i=1}^{n+1} d(i/2)}$$

This permits recursive bootstrapping:

$$d\left(\frac{1}{2}\right) = \frac{1}{1 + y(1/2)/2}$$

$$d\left(\frac{n+1}{2}\right) = \frac{2 - y\left(\frac{n+1}{2}\right)\sum_{i=1}^{n} d(i/2)}{2 + y\left(\frac{n+1}{2}\right)}$$

- Workflow: reference yield + key-rate perturbations → new yield curve → new discount factors → new deterministic-cash-flow price.

A 30-year annuity and its key-rate hedge

- Annuity: \$3,250 every six months for 30 years. Reference par-coupon curve flat at 5%, hence spot curve flat at 5%. $\lambda = 1/(1 + 0.05/2) = 0.975609756$.

Initial price:

$$P = 3{,}250\sum_{i=1}^{60}\lambda^i = \$100{,}453.13$$

Single +1 bp shock	Repriced annuity	Key DV01	Key duration
2y	100,452.15	0.98	0.098
5y	100,449.36	3.77	0.375
10y	100,410.77	42.36	4.217
30y	100,385.88	67.25	6.695

- Parallel +1 bp first-order: $\Delta P = -(0.98+3.77+42.36+67.25) = -114.36 \Rightarrow$ P≈100,338.77. Exact flat 5.01% price=100,338.81.
- Illustration nonparallel move: +5 bp (2y), +3 bp (5y), +2 bp (10y), −1 bp (30y) ⇒ $\Delta P \approx -33.68 \Rightarrow$ P≈100,419.45.

Hedge the annuity with par bonds

Par bond maturity	DV01 per 100 face	Duration
2y	0.018810	1.8810
5y	0.043760	4.3760
10y	0.077946	7.7946
30y	0.154543	15.4543

- Match each annuity key-rate DV01 separately. Required faces:
  $F_{2y} = \$5,209.99;\ F_{5y} = \$8,632.93;\ F_{10y} = \$54,345.32;\ F_{30y} = \$43,515.40.$

Bucket shifts, volatility weighting, and principal components

General curve move:

$$y(t) \to y(t) + \sum_{i=1}^{N} x_i X_i(t)$$

- If sold security price is f(x_1,...,x_N) and hedge portfolio price is g(x_1,...,x_N), choose holdings so $\partial f/\partial x_i = \partial g/\partial x_i$ at the unperturbed curve $x = 0$, for every i. Usually N factors suggest N hedge instruments.
- Bucket shifts: parallel shifts of many forward-rate "buckets"; frequently used to hedge swaps.
- Technical note: key-rate shock shapes are chosen largely for mathematical convenience. Actual term-structure moves do not have to look like those shocks, and real key rates are generally correlated with one another.
- Multifactor hedging lies between simple one-factor hedging and exact cash-flow immunization. Regression-based hedging is another practical multifactor extension.

Volatility-weighted hedging

- If historical moves have different amplitudes across maturities, weight DV01 by typical relative movement rather than assume 1-for-1 parallel shifts.

If 1 bp in 30y typically means 1.1 bp in 20y:

$$F_{20} \cdot 1.1 \cdot \frac{\mathrm{DV01}_{20}}{100} = F_{30} \cdot \frac{\mathrm{DV01}_{30}}{100}$$

- Illustration: short \$10mm face 20y par; yields 5.8% (20y), 6% (30y); DV01s 0.117465 and 0.138378.

  Volatility-weighted hedge buys \$9,337,575 face 30y; parallel-shift hedge would buy \$8,488,704.
- Illustration: hedge short \$10mm 20y with equal face amounts of 10y and 30y. $y_{10} = 5.2\%$, $y_{20} = 5.8\%$, $y_{30} = 6\%$. 1 bp in 30y corresponds to 1.1 bp in 20y and 1.2 bp in 10y. DV01s 0.077215, 0.117465, 0.138378 ⇒ F=\$5,592,700 in each 10y and 30y bond.

PCA

- Principal-component hedging identifies uncorrelated random variables that explain observed term-structure movements. First PC looks much like a parallel shift and captures about 80% of variance; first 3 PCs typically capture about 95%.

A factor representation $P = f(y_1, \dots, y_N)$ gives

$$\Delta P \approx \sum_{i=1}^{N} \frac{\partial P}{\partial y_i} \Delta y_i, \qquad \boxed{\mathrm{KRDV01}_i = -10^{-4} \frac{\partial P}{\partial y_i}}.$$

One-factor duration retains one direction; a key-rate vector retains selected maturity shocks; bucket hedges use grouped forward-rate shocks; volatility weighting adjusts for typical movement size; PCA uses empirically estimated orthogonal factor directions. Complete cash-flow matching is a stronger replication criterion than matching finitely many derivatives.

Part XII — Hedging, Immunization, and Relative-Value Strategies

After exposures have been measured, hedging is a linear-algebra problem: choose instrument positions so the resulting sensitivity vector is zero or deliberately shaped.

DV01, duration, and convexity neutralization

For a target exposure A hedged using h units of a traded instrument B, first-order neutrality requires

$$\boxed{\mathrm{DV01}_A + h\,\mathrm{DV01}_B = 0 \quad \Longrightarrow \quad h = -\frac{\mathrm{DV01}_A}{\mathrm{DV01}_B}.}$$

The units must first be made consistent (per face, per contract, or per full position). For multiple constraints, construct the hedge matrix from dollar-duration, dollar-convexity, or key-rate exposures and solve $\mathbf{Hh} = -\mathbf{r}$. Matching only duration leaves quadratic-rate exposure; matching both duration and convexity addresses the first two local terms but not all curve scenarios.

Cash-flow immunization versus factor-based hedging

Exact cash-flow immunization replicates every promised payment at each relevant date and state; under the stated frictionless no-default assumptions, the hedged net cash flow is identically zero. By contrast, a DV01, key-rate, bucket, or PCA hedge cancels sensitivities only to selected small movements. The latter generally uses fewer securities and less balance-sheet capacity, but retains unhedged curvature, basis, and model risk.

Bullet, barbell, and butterfly portfolios

- Barbell: combine short- and long-maturity bonds instead of an intermediate-maturity "bullet." A barbell can be constructed to match the bullet's price and duration but generally has higher convexity on a flat curve.
- Illustration, flat yield 6%: 9-year ZCB bullet has D_{Mac}=9 and C=80.59195. Barbell 75% in 2-year zeros +25% in 30-year zeros also has D_{Mac}=9 but C=219.15355.
- Therefore under parallel shifts away from 6%, the barbell price is above the bullet price for both rate rises and falls. First-order approximations are identical; second-order captures the difference and is close to exact repricing.
- This is not an arbitrage in practice: the real curve is not flat and changes need not be parallel. Example nonparallel curve $\hat{r}(2) = 6.2\%$, $\hat{r}(9) = 6.4\%$, $\hat{r}(30) = 6.6\%$ gives bullet 96.5685 vs barbell 95.7063 — bullet outperforms.

General flat-curve proof:

$$g(t) = \frac{t(t+0.5)}{(1+y/2)^2} \quad \text{is strictly convex}$$

$$C_{\mathrm{barbell}} = \alpha g(T_1) + (1-\alpha) g(T_2) > g(\alpha T_1 + (1-\alpha)T_2) = C_{\mathrm{bullet}}$$

- Butterfly trade: buy/sell an intermediate maturity while selling/buying two maturities that straddle it.
- also emphasizes **steepener** and **flattener** trades. These are relative-value curve trades, so their risk should be understood with maturity/key-rate exposures rather than with a single parallel-shift DV01 alone.

If two portfolios match initial price and duration, their second-order response depends on convexity. A barbell often has greater convexity than a bullet, but its exposures concentrate at different

maturities; a steepening or flattening curve can overturn the parallel-shift ranking. A butterfly deliberately builds positive and negative positions at several tenors to isolate curve curvature, subject to the precise hedge constraints chosen.

Volatility-weighted, key-rate, and PCA hedges

Volatility weighting replaces an implausible one-basis-point-to-one-basis-point co-movement assumption with estimated relative factor changes. Key-rate hedging neutralizes specified curve nodes; PCA hedging neutralizes dominant empirically estimated movements. In either method the correct object is a *signed portfolio sensitivity vector,* rather than the number of long or short bonds.

Part XIII — Real Markets: Rates, Government Securities, Futures, and Credit

The framework finally meets the conventions that determine actual quoted instruments, collateral, funding, counterparty exposure, and the meaning of different yield curves.

Market participants, trading conventions, and risk taxonomy

Products, rates, market participants, and risks

- Products listed: ZCBs; coupon bonds; annuities; TIPS; floaters/inverse floaters; callable/putable bonds; swaps, caps, floors, swaptions; SOFR futures; bond options; bond futures; options on interest-rate/bond futures.
- Interest rates depend on initiation date, loan length, and payment schedule. SOFR (Secured Overnight Financing Rate) is the key benchmark example. Future market spot rates are generally unknown today.
- Deterministic-cash-flow securities can be studied extensively without an interest-rate evolution model. Securities with uncertain payments generally require a mathematical model of rate uncertainty.

Market role	Examples
Issuers	Federal/state/local governments, agencies/GSEs, corporations
Intermediaries	Dealers, investment banks, credit-rating agencies
Investors	Individuals, governments, insurers, pension funds, mutual funds, commercial banks

Risk	Meaning / Meaning
Default risk	Issuer/counterparty may fail to pay; can sometimes be hedged with CDS.
Inflation risk	Real value of nominal payments changes.
Liquidity risk	Difficulty/cost of trading; important to traders.
Interest-rate (market) risk	Prices change when rates move; central topic.
Reinvestment risk	Interim cash flows may be reinvested at unknown future rates.
Currency (FX) risk	Exchange-rate exposure.
Timing/call risk	Cash-flow timing may change due to embedded options; the only listed risk that is not relevant to truly deterministic cash flows.

U.S. Treasury securities, on-/off-the-run markets, and STRIPS

Security	At-issue maturity / payment convention
T-Bills	Zeros: 4, 6, 8, 13, 17, 26, or 52 weeks
T-Notes	2, 3, 5, 7, or 10 years; semiannual coupons
T-Bonds	20 or 30 years; semiannual coupons

TIPS	5, 10, or 30 years; inflation protected; semiannual coupons
FRNs	2-year; quarterly floating coupons
STRIPS	Individual coupon or principal pieces: C-strips and P-strips

- Representative Treasury issuance cadence: bills <1 year weekly; 1-year bills every 4 weeks; 2/3/5/7-year notes monthly; 10-year notes and 20/30-year bonds four times/year.
- Most recently issued security of a maturity = on-the-run; older issues = off-the-run. On-the-run securities are typically much more liquid.
- STRIPS = Separate Trading of Registered Interest and Principal Securities. Stripping creates long-maturity zeros. To reassemble a bond, coupon payments must use C-strips and principal a P-strip. Any correct-date/amount C-strip works, but the P-strip must have the same CUSIP as the bond; hence same-date/amount C- and P-strips can trade at different prices.
- **TIPS**: face value is periodically adjusted to F_{adj} using CPI-U. The fixed annual coupon rate q is applied to adjusted principal, so each semiannual coupon is $F_{\text{adj}}q/2$. Deflation can reduce adjusted principal during the life of the bond, but maturity principal is protected at $\max\{F, F_{\text{adj}}\}$.
- **Treasury FRNs**: 2-year notes paying quarterly coupons. The coupon rate is the 3-month T-bill rate plus a spread fixed at auction; the floating reference rate resets weekly while interest is paid quarterly.

Benchmark rates, SOFR, federal funds, repo, and haircuts

- **LIBOR** was an unsecured bank funding benchmark quoted for several terms, commonly on an actual/360 basis. USD LIBOR ceased to be quoted on June 30, 2023. a historical fallback for legacy 3-month LIBOR contracts was 3-month SOFR + 26.161 bp unless another fallback was negotiated.
- **SOFR** (Secured Overnight Financing Rate) is a broad measure of the cost of overnight cash borrowing collateralized by U.S. Treasuries and is the preferred U.S. alternative to LIBOR. Any numerical SOFR quotation is historical context, not a timeless model input.
- **Fed funds**: banks lend/borrow reserve balances overnight in the federal funds market; the Federal Reserve publishes the effective federal funds rate and targets this market as part of monetary policy. The rate influences many other short rates.
- Modern fixed income uses multiple economically distinct rates: Treasury yields, Fed Funds, repo, SOFR, legacy LIBOR, and other bank-credit-sensitive benchmarks. Regulation and collateral matter, so there is not one universal observed “risk-free rate” for every transaction.
- **Repo**: the securities owner sells collateral and agrees to repurchase it later at a higher price; economically this is a collateralized loan. For the borrower it is a repo; for the lender it is a reverse repo. Overnight repo is most common, though term repos also exist. Treasury securities are common collateral.
- **Haircut**: because collateral value can fall, the lender can require collateral worth more than the cash lent. The excess collateral is the haircut.

Exchange-traded SOFR and Treasury futures, conversion factors, and CTD

Exchange-traded rate and Treasury futures — market conventions

- Eurodollar futures are being replaced by SOFR futures. The illustrative contract specifications include: one 3-month SOFR futures contract has $1,000,000 notional and DV01 $25; one 1-month SOFR futures contract has $5,000,000 notional and DV01 $41.67. The 3-month contract rate compounds daily SOFR; the 1-month contract uses an arithmetic average of daily SOFR.

- Treasury note/bond futures are based roughly on a hypothetical 6% coupon bond. Multiple Treasury securities may be acceptable for delivery, so conversion factors adjust their deliverable face equivalence. As delivery approaches, one security typically becomes the **cheapest to deliver (CTD)**.

Contract	Face amount per contract
ZT (2y), Z3N (3y)	$200,000
ZF (5y), ZN (10y), TN (ultra 10y)	$100,000
ZB (30y), UB (ultra 30y)	$100,000

The Treasury-futures prices used here are quoted per $100 face.

Credit spreads, structural default, credit protection, and OAS

Credit risk, OAS, and multiple curves — context

- A **credit spread** is the yield difference between a defaultable bond and a U.S. Treasury with the same promised payment structure. If no directly comparable Treasury exists, especially for a bond with embedded options, an option-adjusted spread is more useful.
- If an interest-rate model is calibrated to the Treasury spot curve, the **OAS** is the shift applied to that Treasury curve that makes $P_{\text{Model}} = P_{\text{Market}}$. OAS is model-dependent: different rate models can produce different OAS values for the same security.
- Two broad default-model classes are **structural models**, which model the firm and economic cause of default, and **reduced-form models**, which use observed market/default statistics. In practice, many institutions use reduced-form models.
- **Merton model**: a firm has total value $F(t)$ and zero-coupon debt of face K due at T. At maturity, bondholders receive $\min\{F(T), K\}$ and equity receives $(F(T) - K)^+$. Hence equity is a call on firm value with strike K, while risky debt equals a default-free K-face ZCB minus a put on firm value.
- A **credit default swap (CDS)** is economically similar to insurance for a security that may default and is an important default-risk hedge.
- **Multicurve discounting**: after 2008, collateral became much more important. Collateral can earn a rate different from the floating reference rate in a swap, so one should not assume that a single “risk-free rate” can be used for every role in pricing.

Multicurve perspective. A collateralized derivatives contract may be discounted at a rate associated with its collateral agreement, while its floating cash flows refer to another published rate. The projection curve and discount curve therefore need not coincide. This separates funding, forecasting, and credit-risk effects rather than treating every quoted yield as an interchangeable risk-free rate.

Concept Checklist: The Logical Order of the Field

1. **Pricing foundations.** Frictionless-market assumptions; no arbitrage; strong arbitrage; law of one price; replication; discounted cash-flow valuation; zero-coupon building blocks; discount factors; simple, nominal and continuous compounding; annual, semiannual and monthly frequencies; day counts; clean/dirty prices; accrued interest; bill discount and bond-equivalent yields.

2. **Term structure.** Spot rates; forward rates and forward loans; future realized spot rates; par-coupon rate; annuity yield; YTM and IRR; premium/par/discount relations; coupon effect; yield-curve bootstrapping; smooth interpolation; Nelson–Siegel; cash-flow reinvestment.

3. **Basic instruments.** Zero-coupon and coupon bonds; self-amortizing annuities and perpetuities; STRIPS; floating-rate notes; inverse floaters; TIPS; Treasury FRNs; inflation and credit exposure.

4. **Uncertain rates.** State-contingent payoffs; physical versus pricing probability; one-period binomial replication; no-arbitrage restriction; adapted and predictable processes; conditioning; tower property; short-rate processes; stochastic discounting; discounted-price martingales.

5. **Models and algorithms.** Short-rate versus whole-curve models; Ho–Lee; Vasicek; Hull–White; Black–Derman–Toy; calibration versus model dynamics; backward induction; pre-/post-payment conventions; state variable $k = \#H$; recombining trees; conditional pricing of multiple cash flows.

6. **Contingent products.** Rate swaps; swap rates; forward-rate decomposition; caplets and floorlets; cap/floor parity; European, American and Bermudan exercise; swaptions; callable and putable bonds; embedded call and put values; exercise and negative convexity.

7. **Contract timing.** Forward settlement and coupon stripping; cash-and-carry; central clearing; daily futures marking-to-market; martingale futures price; forward–futures covariance basis; reinvestment of margin cash flows.

8. **Risk and hedging.** Yield duration; Macaulay duration; DV01/PVBP; dollar duration; convexity; dollar convexity; portfolio aggregation; finite differences; option kinks; key-rate DV01 and duration; bucket factors; volatility weighting; PCA; complete cash-flow immunization; DV01/convexity hedges; barbell, bullet and butterfly trades.

9. **Market structure.** SOFR; federal funds; LIBOR legacy conventions; repo/reverse repo; collateral haircuts; Treasury securities and quotation; SOFR/Treasury futures; conversion factors; cheapest-to-deliver; structural and reduced-form credit; Merton equity-as-call; credit spread; CDS; OAS; multicurve discounting.

Formula Checklist: From Cash Flow to Risk

Deterministic cash-flow valuation

$$P = \sum_i CF_i d(t_i), \quad P_{\text{ZCB}} = Fd(T), \quad P_{\text{bond}} = Fd(T) + \frac{Fq}{2}\sum_{i=1}^{2T} d(i/2)$$

Accumulation and rate conventions

$$d(t) = (1 + r^{(m)}(t)/m)^{-mt} = e^{-tr_c(t)}, \quad \hat{r}(t) = 2\{d(t)^{-1/(2t)} - 1\}$$

Spot and forward rates

$$\left(1 + R^{\text{for}}(0,\eta,T)\right)^{T-\eta} = \frac{d(\eta)}{d(T)}, \quad f(t) = 2\left(\frac{d(t-1/2)}{d(t)} - 1\right)$$

Par yields and par bootstrapping

$$q_n = \frac{2(1-d(t_n))}{\sum_{i=1}^{n} d(t_i)}, \quad d(t_n) = \frac{1 - (q_n/2)\sum_{i=1}^{n-1} d(t_i)}{1 + q_n/2}$$

YTM and price classification

$$P = \sum_i CF_i(1 + y/2)^{-2t_i}, \qquad P \gtrless F \iff q \gtrless y$$

Annuities and perpetuities

$$P_A = A\sum_{i=1}^{2T}(1 + y/2)^{-i}, \qquad P_\infty = 2A/y$$

Duration and DV01

$$D = -\frac{P'(y)}{P}, \quad \text{DV01} = -\frac{P'(y)}{10000} = \frac{PD}{10000}, \quad D_\$ = PD$$

Macaulay and convexity

$$D_{\text{Mac}} = \sum_i w_i t_i = (1 + y/2)D, \quad C = \frac{P''(y)}{P} = \frac{\sum_i w_i t_i (t_i + 1/2)}{(1 + y/2)^2}$$

Second-order price change

$$\frac{\Delta P}{P} \approx -D\Delta y + \frac{1}{2}C(\Delta y)^2$$

Portfolio aggregation

$$\text{DV01}_p = \sum_i \alpha_i \text{DV01}_i, \quad D_p = \sum_i \frac{\alpha_i P_i}{P_p} D_i, \quad C_p = \sum_i \frac{\alpha_i P_i}{P_p} C_i$$

Key-rate exposures

$$\Delta P \approx \sum_i \frac{\partial P}{\partial y_i}\Delta y_i, \quad \text{KRDV01}_i = -\frac{1}{10000}\frac{\partial P}{\partial y_i}$$

Risk-neutral probabilities

$$\tilde{p} = \frac{1 + R_0 - d}{u - d}, \quad \tilde{q} = \frac{u - 1 - R_0}{u - d}, \quad d < 1 + R_0 < u$$

Stochastic discounting

$$D_n = \prod_{j=0}^{n-1}(1+R_j)^{-1}, \quad V_n = D_n^{-1}\widetilde{\mathbb{E}}_n[D_m V_m]$$

One-step recursion

$$V_n = \frac{\tilde{p}V_{n+1}^H + \tilde{q}V_{n+1}^T}{1+R_n}, \quad V_n = \frac{\widetilde{\mathbb{E}}_n[V_{n+1}+C_{n+1}]}{1+R_n}$$

Model zero and floating notes

$$B_{n,m} = D_n^{-1}\widetilde{\mathbb{E}}_n[D_m], \quad B_{n,n+1} = (1+R_n)^{-1}$$

Swap par rate

$$\mathrm{SR}_{0,m} = \frac{1-B_{0,m}}{\sum_{i=1}^{m} B_{0,i}}$$

Cap–floor parity

$$\underset{\substack{\text{Receive-fixed}\\ \text{swap}}}{(K-R)} + \underset{\text{Cap}}{(R-K)^+} = \underset{\text{Floor}}{(K-R)^+}$$

American and Bermudan exercise

$$V_m = G_m, \quad V_n = \max\left(G_n, \frac{\widetilde{\mathbb{E}}_n[V_{n+1}]}{1+R_n}\right) \quad (n \text{ exercisable})$$

Callable and putable bonds

$$V_n^c = \min\left(F_n, \frac{\widetilde{\mathbb{E}}_n[V_{n+1}^c + Fq]}{1+R_n}\right), \quad V_n^p = \max\left(F_n, \frac{\widetilde{\mathbb{E}}_n[V_{n+1}^p + Fq]}{1+R_n}\right)$$

Forwards and futures

$$\mathrm{For}_{n,m} = \frac{\widetilde{\mathbb{E}}_n[D_m P_m]}{\widetilde{\mathbb{E}}_n[D_m]}, \quad \mathrm{Fut}_{n,m} = \widetilde{\mathbb{E}}_n[P_m]$$

Cash-and-carry and covariance

$$\mathrm{For}_{n,m} = P_n/B_{n,m} \text{ (no interim payments)}, \quad \mathrm{For}_{0,m} - \mathrm{Fut}_{0,m} = \frac{\mathrm{Cov}_{\widetilde{\mathbb{P}}}(D_m, P_m)}{\widetilde{\mathbb{E}}[D_m]}$$

The Fixed Income Discipline Map

NO ARBITRAGE

↓

Law of One Price / Replication

↓

VALUE OF CASH FLOWS $d(t), D_n, \widetilde{\mathbb{P}}$

↓

TERM STRUCTURE Spot ↔ Forward ↔ Par (YTM: security summary)

↓

RATE DYNAMICS Ho–Lee / Vasicek / Hull–White / BDT; Key Rates / PCA

↓

PRICING ENGINE $P = \sum CF\, d(t), \quad V_n = D_n^{-1}\widetilde{\mathbb{E}}_n[D_m V_m]$

Backward Induction

↓

PRODUCTS ZCB / Coupon / FRN / Swap; Forward / Futures

Cap / Floor / Options; Callable / Putable / Swaption

↓

RISK DV01 / Duration / Convexity / KRDV01

↓

HEDGING / STRATEGIES

DV01 / Convexity / Key-Rate / PCA / Immunization

Barbell / Bullet / Butterfly

↓

REAL MARKETS Treasury / SOFR / Repo / Futures

Credit Spread / OAS / CDS / Multicurve

Fixed Income is the study of how cash flows across time and states are priced, how those prices change when the term structure changes, and how that risk is transferred, structured, and hedged through financial instruments.

ABOUT THE AUTHOR

Mizuki Seika

Quantitative Trader · Web3 Explorer · Independent Idol & Music Producer

I'm Mizuki Seika, a quantitative trader working at the intersection of mathematics, financial markets, and emerging technology; an explorer of Web3 and cryptocurrency; and, in a parallel universe that happens to be equally real, an independent idol and original music producer.

My academic path spans Carnegie Mellon University's Master of Science in Computational Finance (MSCF '28), Mathematics at Tsinghua University ('25), and an exchange year at the University of Oxford ('24). My interests range from quantitative research and systematic trading to decentralized finance and digital assets.

To me, markets and music have more in common than they appear to: both reward an ear for signals, a sense of structure, and the courage to build something original.

> *In finance, I turn complexity into models.*
> *In music, I turn it into melodies.*

This work is where I bring that same creative discipline to learning --- reconstructing knowledge until the pieces form a framework that actually makes sense.

Explore My Worlds

LinkedIn
Original music

GitHub
Open-source projects and research

Spotify
Original music

Instagram
The creative side, on and off stage

Thank you for reading.